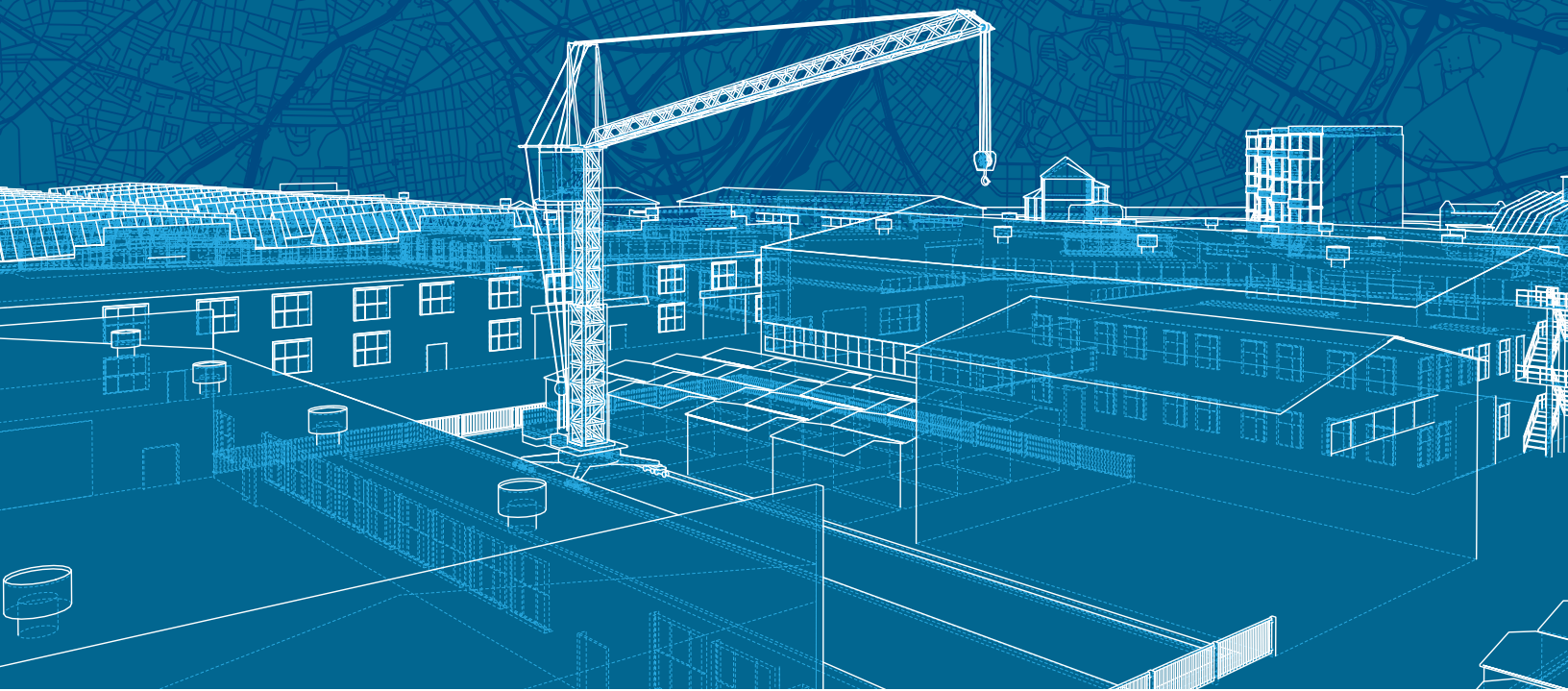


CITIES AS GROWTH HUBS

KNOWLEDGE FRAMEWORK



Ministry of Finance
Department of Economic Affairs
Ministry of Housing and Urban Affairs
Government of India



CITIES AS GROWTH HUBS

KNOWLEDGE FRAMEWORK



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Ministry of Housing and Urban Affairs
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Cities as Growth Hubs Knowledge

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ABBREVIATIONS AND ACRONYMS

ABD	Area-based development	DTCP	Directorate of Town and Country Planning
ADB	Asian Development Bank	DULT	Directorate of Urban and Land Transport
AI	Artificial intelligence	EC	Economic Census
AIIMS	All India Institute of Medical Sciences	ELCOT	Electronics Corporation of Tamil Nadu Limited
AISHE	All-India Survey on Higher Education	EPC	Environmental Planning Collaborative
AMC	Ahmedabad Municipal Corporation	EPZ	Export processing zone
AMRUT	Atal Mission for Rejuvenation and Urban Transformation	ESDM	Electronics system design and manufacturing
ASI	Annual Survey of Industries	ESG	Environmental, social, and governance
ASUSE	Annual Survey of Unincorporated Sector Enterprises	EV	Electric vehicle
AUDA	Ahmedabad Urban Development Authority	FAR	Floor area ratio
BCP	Bhubaneswar-Cuttack-Puri	FIs	Financial intermediaries
BID	Business improvement district	FSI	Floor space index
BPO	Business process outsourcing	FY	Fiscal year
CBUA	Contiguous built urban agglomeration	GCCs	Global Capability Centers
CBS	Current business register	GDCR	Gujarat Comprehensive Development Control Regulations
CMDA	Chennai Metropolitan Development Authority	GDP	Gross domestic product
CEDC	City Economic Development Council	GHG	Greenhouse gas
CFC	Central Finance Commission	GIFT	Gujarat International Finance Tec-city
CFCs	Common Facility Centers	GIS	Geographic information system
CIIP	City infrastructure and investment plan	GLMS	Government Land Management System
CMA	Chennai Metropolitan Area	GTPUDA	Gujarat Town Planning & Urban Development Act of 1976
CMDA	Chennai Metropolitan Development Authority	HHI	Herfindahl–Hirschman index
CMPs	Comprehensive mobility plans	HLC	High-Level Committee
CRUT	Capital Region Urban Transport	HUDCO	Housing and Urban Development Corporation
CSR	Corporate Social Responsibility	IADBs	Industrial Area Development Boards
CSS	Centrally Sponsored Scheme	IIHS	Indian Institute for Human Settlements
CUMTA	Chennai Unified Metropolitan Transport Authority	IIM	Indian Institute of Management
DLF	Delhi Lease and Finance	IISER	Indian Institute of Science and Education
DMIC	Delhi-Mumbai Industrial Corridor	IIT	Indian Institute of Technology
DNCR	Delhi National Capital Region	ILASP	Integrated Land Administration and Spatial Planning Project

ABBREVIATIONS AND ACRONYMS

IPT	Intermediate public transport	NMT	Non-motorized transport
IT	Information technology	ODOP	One District One Product
ITes	IT enterprise solutions	OECD	Organisation for Economic Co-operation and Development
IUO	India Urban Observatory	OSR	Own-source revenue
JNNURM	Jawaharlal Nehru National Urban Renewal Mission	OTOP	One Tambon One Product
KAMCO	Korea Asset Management Company	OVOP	One Village One Product
KCJ	Khordha–Cuttack–Jagatsinghpur	PCPIR	Petroleum, Chemicals and Petrochemical Investment Regions
LBF	Land-based financing	PLFS	Periodic Labor Force Survey
LVC	Land-value capture	PMAY	Pradhan Mantri Awas Yojana
MANIT	Maulana Azad National Institute of Technology	PMR	Pune Metropolitan Region
MCGM	Municipal Corporation Greater Mumbai	PMRDA	Pune Metropolitan Region Development Authority
MCI	Ministry of Commerce and Industry	PPP	Public–private partnership
MDBs	Multilateral development banks	PRC	People’s Republic of China
MIDC	Maharashtra Industrial Development Cooperation	R&D	Research and development
MMLP	Multimodal Logistics Park	RBI	Reserve Bank of India
MMR	Mumbai Metropolitan Region	SASCI	Special Assistance to States for Capital Investment
MMRDA	Mumbai Metropolitan Region Development Authority	SCADA	Supervisory Control and Data Acquisition
MOHUA	Ministry of Housing and Urban Affairs	SCM	Smart Cities Mission
MOSPI	Ministry of Statistics and Programme Implementation	SEBI	Securities Exchange Board of India
MPLIS	Multi-Purpose Land Information System	SEZs	Special economic zones
MRTS	Mass rapid transit system	SFC	State Finance Commission
MSMEs	Micro, small, and medium enterprises	SIPCOT	State Industries Promotion Corporation of Tamil Nadu
MTC	Metropolitan Transport Corporation	SIR	Special Investment Region
MW	Megawatts	SITPE	Scheme for Integrated Textile Parks
NAAC	National Assessment and Accreditation Council	SMEs	Small and medium enterprises
NATRAX	National Automotive Test Tracks	SPV	Special purpose vehicle
NCR	National Capital Region	SREDA	Surat Regional Economic Development Authority
NCRPB	National Capital Region Planning Board	SRFDLC	Sabarmati Riverfront Development Corporation Limited
NICDC	National Industrial Corridor Development Corporation	STP	Sewerage Treatment Plant
NICDP	National Industrial Corridor Development Program	TANSIDCO	Tamil Nadu Small Industries Development Corporation Limited
NITI	National Institution for Transforming India	TAP	Treated Aquifer Project
NIUA	National Institute of Urban Affairs		

TCPO	Town and Country Planning Organization	TOD	Transit-oriented development
TDR	Transferable development right	TPO	Town Planning Officer
TIDCO	Tamil Nadu Industrial Development Corporation Limited	TPS	Town and Country Planning Scheme
TIF	Tax Increment Financing	ULB	Urban local body
TMP	Third Master Plan (Chennai Metropolitan Development Authority)	URDPFI	Urban and Regional Development Plans Formulation and Implementation Guidelines
TNIFNC	Tamil Nadu Infrastructure Fund Management Corporation	USD	United States dollars
TNUDF	Tamil Nadu Urban Development Fund	WBG	World Bank Group
		WRI	World Resources Institute

EXECUTIVE SUMMARY

India's urban population is growing, which could be a boon to development, given the critical role of cities in driving economic growth and creating jobs, especially in the context of India's goal of becoming a \$30 trillion economy by 2047, as envisioned in *Viksit Bharat @2047*. But India's urbanization is underleveraged. Difficulties in accessing affordable serviced land, gaps in transport and other urban infrastructure, a paucity of purpose-fit financing for catalytic infrastructure and projects, unplanned peri-urban expansion and an urban form that does not sufficiently promote livability and resilience contribute to congestion forces that constrain agglomeration economies which drive productivity and innovation in cities. Addressing challenges facing firms and workers could help Indian cities better realize their potential to serve as effective growth hubs. The Government of India in its FY 24-25 budget highlighted "cities as growth hubs" as a key priority, to be achieved by "economic and transit planning, and orderly development of peri-urban areas utilizing town planning schemes." This report has been prepared in response to a request from the Department of Economic Affairs and the Ministry of Housing and Urban Affairs, Government of India, to develop guidance on this topic.

Specifically, this report offers a framework to support Indian cities in harnessing their potential as urban growth hubs and proposes a roadmap for implementing it. Urban growth hubs are cities and urban areas that are competitive, livable, inclusive, and resilient. Not all cities, however, are urban growth hubs. They must have economic potential—namely, capacity for igniting and sustaining growth and creating jobs which depends on key enablers such as access to skills, entrepreneurship, markets, connectivity, and institutional capacity. The proposed framework builds on urban planning reforms being carried out by the Ministry of Housing and Urban Affairs, notably through its "Special Assistance to States for Capital Investment" scheme; the recommendations of the High-Level Committee on Urbanization to envision the role of cities as gateways to enable better trade and economic activities, and Niti Ayog's Growth Hubs program under *Viksit Bharat @2047*, to develop urban areas in a structured way.

FRAMEWORK FOR PROMOTING URBAN GROWTH HUBS

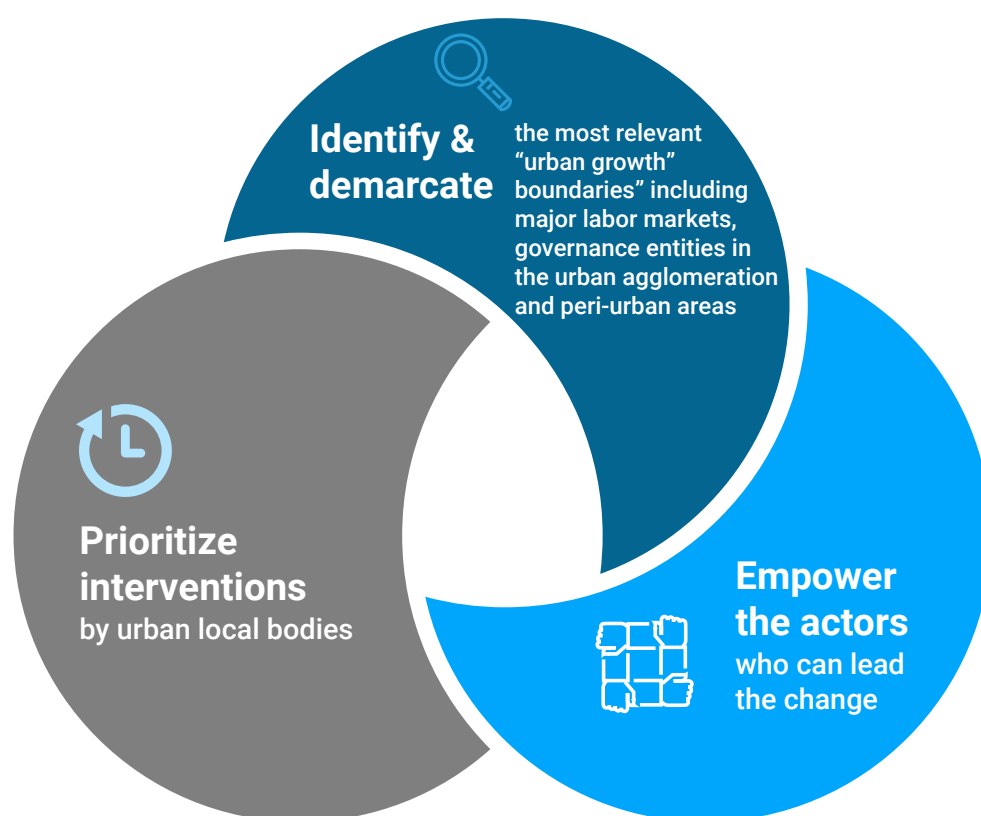
For programs to be responsive to the specific spatial contexts, growth drivers, constraints to economic activity, and the economic potential of cities in India's complex city system, which comprises of 4,041 statutory cities and towns as per the 2011 census, a one-size-fits-all approach to urban growth will not work and a new approach to incentivizing urban growth hubs is needed—for several reasons.

- First, satellite analysis undertaken as part of this report shows that cities across class sizes face unplanned peri-urban growth, and in 2020, an estimated thirty two percent of built-up area was outside of statutory city boundaries. Cities are integrated labor markets, but due to unplanned peri-urban growth, the geographic boundaries of local labor markets do not necessarily coincide with those of existing administrative units and municipal boundaries. As such cities need to rethink their "urban growth boundaries."
- Second, cities face substantial differences in governance structures, municipal capacities, access to markets, connectivity, skills, and labor pool. Thus, the scope for growth varies across urban areas.
- Third, most cities face many cross-cutting constraints such as access to serviced land and infrastructure finance and will need to prioritize interventions.
- Fourth, while industrial estates, corridors and zones are critical in terms of providing employment opportunities, systematic planning for urbanizing areas around industrial zones lags industrial planning. Strengthening India's system of cities will be critical to meeting its ambitious industrial and economic development goals.

The proposed framework (figure ES.1) suggests that city administrations will need to take the following key actions:

- Identify and demarcate the most relevant “urban growth boundaries” and plan for it in an integrated way to better address challenges relating to peri-urban growth—a key concern raised in the FY 24-25 budget statement.
- Engage and empower relevant institutional agencies and actors who can lead this change. This includes designating focal entities for undertaking (i) integrated urban, transport and economic planning, and for (ii) economic visioning and planning for the city/urban area. It also involves embedding firm and worker related considerations squarely into urban planning and promoting growth coalitions to crowd in private sector finance.
- Prioritize interventions and investments with a focus on the relevant spatial boundaries given limited resources and capacity at most urban local bodies.

Figure ES.1. Key pillars of the urban growth hub knowledge framework



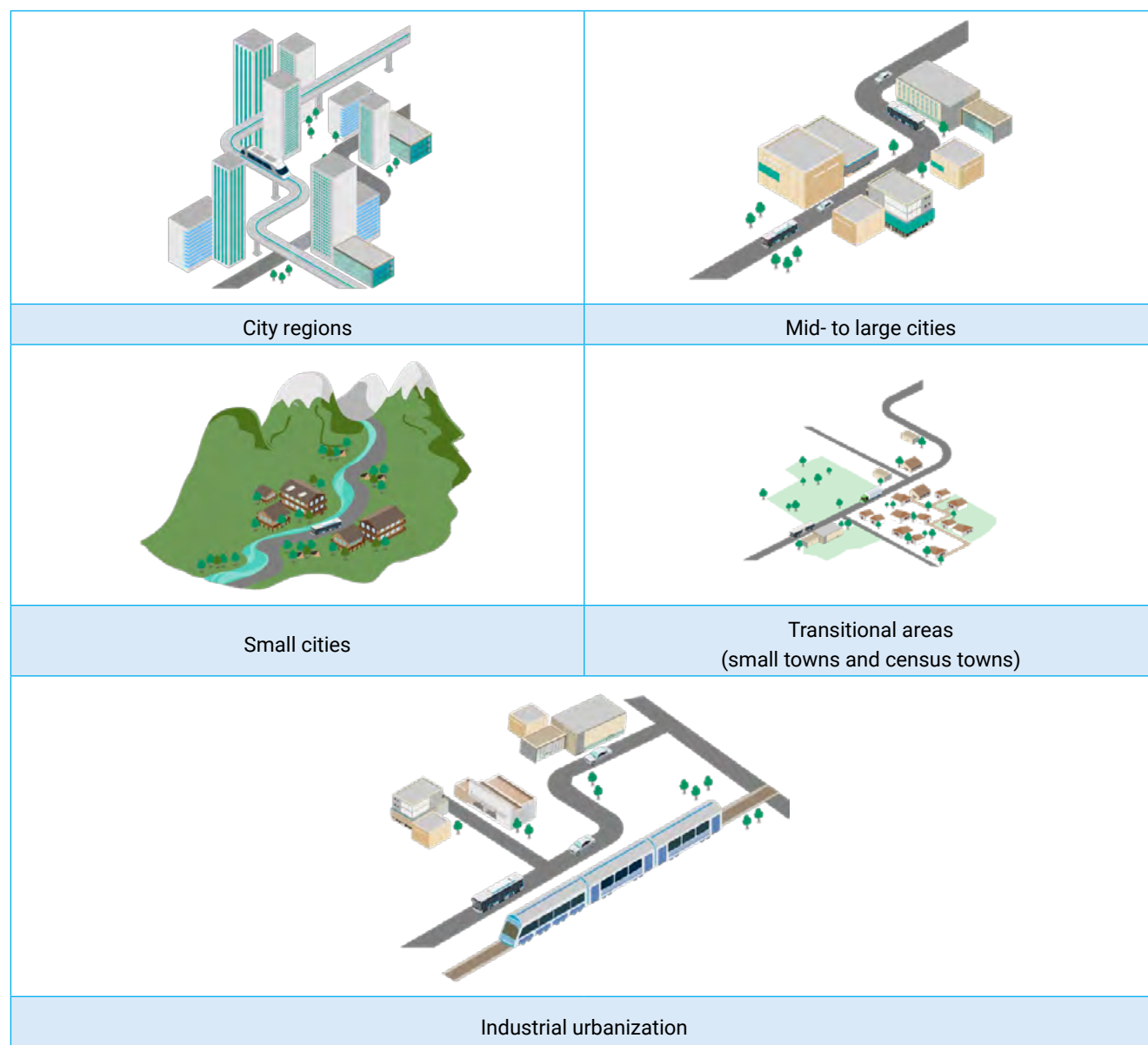
Source: Team Analysis, 2025.

National and international experience shows that there is no “one-size-fits-all” approach to urban growth hubs. Rather, a typology of cities is needed to take the strategy to scale (figure ES.2). Based on India’s city system, five different categories of urban areas are proposed. These include: “city regions” (HLC, 2023; Niti Ayog, 2023); “mid- to large cities”; “small cities”; “transitional areas,” which include small towns and census towns;¹ and “industrial urbanization.” While

transitional areas are not growth hubs, special programs can be put in place to support them.

With this approach to incentivizing urban growth hubs, cities can devise their own growth pathways and strategies tailored to their urban context. Collaboration between cities within each typology over time, can contribute to learning and innovation between cities, and identify common strategies and

Figure ES.2. Typologies of urban areas



Source: Team Analysis, 2025.

growth pathways for specific typologies. Each will need to devise tailored growth strategies.

For city regions and mid-to large cities—such as the Kolkata Metropolitan area comprising of 423 governing entities or Coimbatore urban agglomeration comprising of 186 governing entities—an integrated economic, urban, and transport planning approach is needed, encompassing a clearly demarcated “urban growth boundaries” including both

municipal and peri-urban areas, using planning approaches such as transit-oriented development, that reduces the costs of doing business within the urban agglomeration and enhances resilience and livability. For small cities with unique economic potential and limited economic diversity, such as Ooty (also known as Udhagamandalam), the priority could be on improving the value chain of key economic sectors (such as tourism), improving urban planning and services, and enhancing environmental conservation

within their existing administrative boundaries. For urbanizing areas around well-located industrial estates, special economic zones, and industrial corridors, the priority must be to ensure that those urbanizing areas are systematically planned so that unplanned peri-urban development is mitigated, and agglomeration economies are better leveraged. To facilitate this, high-level focus on this issue including both the Ministry of Housing and Urban Affairs and the Ministry of Commerce and Industry (MCI) will be important.

An important economic development strategy for cities in proximity regardless of typologies is enhancing the interlinkages between a system of neighboring cities. The government of Odisha's efforts at setting up a high-level regional committee to promote interconnectivity between Bhubaneswar, Cuttack, Puri, and Paradip is a model that can be scaled up to other city clusters. Creative brownfield redevelopment is another important strategy and a bottom-up approach that cities can use to revitalize declining urban areas—such as slums, old heritage sites, and urban rivers—through community engagement, local area planning, and land value capture mechanisms to incentivize economic growth and jobs in cities.

CROSS-CUTTING ENABLERS FOR REALIZING THE ECONOMIC POTENTIAL OF CITIES

A review of India's urbanization patterns and experience from urban growth hubs in India and internationally highlight six cross-cutting challenges facing cities. States and cities will have to prioritize which ones are the most pressing in developing their growth strategies.

Strengthening urban governance, integrated planning, and institutional capacity

For cities to be stewards of job growth and economic development, underlying governance issues must be addressed. These include actions at multiple levels. At the central level, a key action includes reforming the Town and Country Planning Office that can work with states to promote a systematic approach to master planning, management of peri-urban growth, transit-oriented development, and climate change.

At the state level, a key governance action includes designating and empowering an entity such as development authorities, urban local bodies or sector agencies to undertake integrated urban, transport, and economic planning for the "urban growth boundary" that includes municipal and peri-urban areas. Also required is filling vacant positions

and prioritizing training and recruitment of urban planners at state and municipal levels, scaling up staffing in other key technical areas such as transport planning, economics, finance, and climate change in state and city sectoral departments (HLC, 2023), and systematically applying a three-tiered system of regional, city, and local area planning for city regions and mid-to large cities and a two-tiered planning system for smaller cities and towns. Resolving these underlying governance issues is critical for enhancing livability and realizing the economic potential of cities.

Designating focal entity for economic visioning, planning, and management

Each of India's cities, like counterparts in many of the world's most dynamic economies, should have a well-staffed and well-resourced entity that carries out economic visioning, planning, and management functions. Working closely with businesses, local and state governments, urban development authorities, and others, the entity would be responsible for developing an economic vision for the city, undertaking regular assessments of the local economy, identifying major bottlenecks and enablers for businesses, leading the simplification of business regulations, identifying financing opportunities, and working closely with the regional/city planning authority to develop strategic plans for economic growth of the urban agglomeration. The entity would lead in assessing skills gaps in priority sectors and in supporting capacity development, in collaboration with universities, to support job growth. An existing institution could assume these responsibilities, where feasible, and its capacities could be built over time. Otherwise, a new institution, such as the city economic development council (HLC, 2023) could be created that is closely integrated with existing city-level governance and planning structures.

Ensuring an adequate supply of affordable serviced land, housing, and productive spaces

Indian states and cities should prioritize enabling adequate supply of serviced land and affordable housing. First, state governments should prioritize developing a modern and efficient land administration system to maintain accurate and updated records of rights and boundaries, which serve as the foundation for transparent and efficient land markets. Inventory of public lands including vacant land should be established. Additionally, land-use conversion should be streamlined to alleviate a critical constraint for firms seeking

EXECUTIVE SUMMARY

to establish or expand their business operations and households seeking affordable housing.

Second, cities should update spatial planning to guide the sustainable supply of serviced land and facilitate locational choices of firms and families. This includes moves to relax restrictive zoning regulations and introduce higher floor space indexes, commensurate with prospective infrastructure upgrades and transit-oriented development principles. As urban areas expand and key infrastructure is developed, state governments must ensure that land acquisition processes are transparent and inclusive. They should advance land pooling by enacting enabling legislation tailored to local contexts, by establishing conflict-resolution mechanisms, and by reducing execution delays.

Third, city governments must empower the private sector to deliver diverse housing solutions for evolving demand by removing barriers to the formal delivery of affordable housing, coordinating land use with infrastructure development, and upgrading and integrating informal housing into the overall market. By leveraging support from, for instance, the Pradhan Mantri Awas Yojana, cities can tackle supply-side constraints such as restrictive land use and zoning regulations and address the affordable housing gap, particularly for economically weaker areas and low-income groups, a significant portion of whom are the young and migrant workers driving urban economies.

Implementing solutions for efficient and sustainable connectivity

Transport supports cities, firms, and economic clusters by facilitating economic concentration and smooth flows of people and goods, essential for urban growth. To support firms, cities must enhance their infrastructure and services, aligning capacity with demand and offering quality inter- and intracity transport services. Cost-effective mobility options should prioritize gender integration, universal accessibility, and climate resilience to create a more inclusive and sustainable transport system.

Cities need to “reinvent” urban spaces by enabling the concentration of jobs and housing in areas with high public transport connectivity through mixed-use development and higher floor space indexes, by improving transport services to reduce reliance on private vehicles, and by promoting efficient use of public transport. Existing and upcoming metro transit corridors present opportunities for urban rejuvenation

through transit-oriented development. High-capacity systems on trunk routes for passengers and freight, alongside flexible systems like buses with minimal access/egress distances and streets fostering walkability, will improve job accessibility. The infrastructure built should have regular resource allocations to support its entire life cycle, and services should have regular funding to maintain service delivery at the targeted level of services. Finally, addressing constraints faced by women in mobility, especially in public transport services, will enhance their access to jobs and increase female labor force participation.

Operationalizing such networks of infrastructure and services requires policy and planning measures, institutional reforms, robust legislative support, and alignment of transport and logistics plans with the statutory development plans/master plans. Integrating freight and passenger planning, developing a common data platform, identifying coordination focal points, and mandating a hierarchy of plans will foster a coordinated approach to transport planning and urban growth, making implementation easier.

Financing critical infrastructure

To generate financing for critical and catalytic infrastructure projects aimed at economic development, cities need to look beyond fiscal transfers, which often restrict their autonomy, and focus on alternative sources, such as land-based financing tools and commercial financing (including private sector participation through public–private partnerships). This shift will require structural and policy reforms at many stages.

First, a conducive regulatory environment for municipal borrowing is needed at the national and state levels to enable cities’ access to capital markets and to private financing. A standardized debt management/capacity assessment framework to support cities in assessing their debt-servicing capabilities will be useful. For smaller cities with potential for economic development, states could establish or strengthen municipal financial intermediaries for pooled financing and technical support.

Second, it is vital to ensure that cities have adequate capacity to absorb bigger volumes of funds and implement larger infrastructure projects. For this, they need access to technical skills to plan, develop, and execute a pipeline of projects geared towards local economic development. Third, administrative and policy-level moves towards strengthening own-source revenues, along with improved governance and

institutional systems, must be targeted to ensure investor confidence and foster private sector participation in infrastructure financing and development. Standardizing policies and financial information, with timely disclosure and progressive taxation, is a good starting point.

Urban amenities, resilience, and livability

Building climate resilience into the heart of a city's urban growth strategies can attract firms by demonstrating a proactive approach to mitigating climate change risks, creating new market opportunities related to sustainability solutions, reducing risks to supply chains, and appealing to a growing pool of talent who prioritize sustainability when choosing where to live and work. Climate resilience needs to be integrated into every aspect of city planning, budgeting, and management, which requires a fundamental shift in governance structures, processes, and practices, as well as strengthened institutional systems. This move requires integrated multisectoral solutions ranging from investments in resilient infrastructure and nature-based solutions, to sustained efforts to strengthen the technical and management capacity of cities.

Cities also need to undertake dynamic and comprehensive climate and disaster-risk assessments to better understand evolving risks spatially, and to put in place early warning systems and emergency response planning, which are investments in saving lives and preserving property. Cities should adopt urban forms and designs that reduce greenhouse gas emissions and make them more resilient, through "blue" and "green" infrastructure that addresses urban heating, flooding, and drought, and that moves from car-centric to walkable cities. States should develop transit-oriented development guidelines and enforce pedestrians' "right to walk," as has been done in the state of Punjab. Cities will also need to support low-income citizens, who are the most vulnerable to disaster and climate risks.

IMPLEMENTING THE URBAN GROWTH HUB FRAMEWORK

This note proposes a seven-step template that cities can use to undertake assessments, spatially demarcate their economically relevant urban areas, namely, "urban growth boundaries," develop a long-term urban growth vision, strategy, and investments, and identify and prioritize options for financing. States will need to designate focal entities for undertaking (i) integrated urban, transport and economic planning, and (ii) economic visioning and planning to move forward and implement the framework. Case studies from Chennai, Bhubaneswar, and Indore highlight how cities can chart their growth path in innovative ways.

PRIORITIZING URBAN GROWTH HUBS

Since a national program could be instrumental in enabling Indian cities to realize their economic potential, this report describes how cities could be selected for support. It proposes prioritizing 100 cities based on their share of the nation's working population—a proxy for cities' contribution to gross domestic product, absent such economic data. It also includes at least one city in each state and covers all city types, thus balancing the goals of promoting productivity and equity. This option is aligned with the idea that economic growth is best sustained when all city typologies function well. The program would be phased over a ten-year time frame and be built up gradually starting with twenty to twenty-five cities, to allow learning by doing and adjustments.

The selection of states and cities could have filters, such as those related to the economic potential of selected cities and the commitment to undertake reforms related to the building blocks: integrating economic, urban, and transit planning; addressing regulatory barriers to land; and enhancing own-source revenue.

EXECUTIVE SUMMARY

KEY ACTIONS AND RECOMMENDATIONS

Table ES.1. Summary of key recommendations

Key issue	Recommendation	Responsible agency	Time frame
Central level actions			
Existing planning guidelines require revision.	Update Model Building byelaws and Urban and Regional Development Plans Formulation and Implementation Guidelines (URDPFI) to address integrated urban, transport and economic planning and sustainability.	MOHUA	ST
Need for higher level action to support states and cities on urban and regional planning.	Re-purpose Town and Country Planning Organization (TCPO) to create a hub which can provide expertise / handholding support to cities (similar to “Urban Renaissance Agency” in Japan). This hub should also have adequate representation from relevant Ministries (like Railways, Roads, Commerce and Industries, Ports etc.).	MOHUA	MT
Growth strategies for different types of cities need to be developed and action incentivized.	Set up a program including funding to support urban growth hubs and develop cross learning between city typologies on economic growth strategies.	MOHUA	MT
Centralized data repository for cities needed.	Set up high quality, regularly updated and accessible national database on cities.	MOHUA	MT
Unplanned urbanization around industrial areas; urban settlements inadequately integrated into industrial planning.	Develop program on “Industrial Area Urban Planning.” Hand holding support to be provided through reformed TCPO. Harness knowledge through partnerships with centers of excellence and education institutions.	MOHUA and MCI	ST-MT
State/city level actions			
Develop typology specific strategies and investment priorities			
Urban peripheral growth; mismatch between administrative and labor markets	Demarcate administrative areas most relevant to the city/relevant urban agglomeration, namely “urban growth boundaries.”	State/ Cities	ST
	Put in place rural-urban transition policy (Odisha good example)	States	MT
ULBs have narrow mandates and capacity; key decisions lie with line departments; thin technical capacity and tax base	Identify and empower key actors to lead the change specifically for (i) integrated urban, transport and economic planning, and (ii) economic visioning and planning; strengthen growth coalitions.	State/ Different City Typologies	ST-MT
Variation in capacity, resources growth drivers (e.g. access to markets, skill base, connectivity) between city typologies	Prioritize interventions and investments with a focus on relevant “urban growth boundaries.”	Different City Typologies	MT

Table ES.1. Summary of key recommendations (cont.)

Key issue	Recommendation	Responsible agency	Time frame
Address most critical cross-cutting constraints			
Governance, planning, and institutional capacity			
Limited availability of trained urban planners at state/city levels	Fill priority vacant positions; scale up training and certification of urban planners	States in partnership with educational institutions	ST
Absence of integrated economic, urban, spatial, and transport planning at metropolitan and city levels	Designate existing or new department to lead integrated economic, urban, transport and logistics planning for metropolitan areas, including peri-urban areas; allocate staffing and budget. Systematize master planning: three tiers for city regions and mid-to large cities and two-tier planning for smaller cities.	States in collaboration with city development authorities and municipal corporations	ST
Limited availability of official data on cities resulting in challenges in measuring gaps, setting service standards, and putting in place performance management systems	Establish state-level multi-sectoral database on urban systems (including spatial, economic, demographic, land, resilience and other data)	States	MT
Strengthening cities' institutions on economic development			
Absence of city-level focal department for economic visioning and planning	Designate existing agency/or establish City Economic Development Council (CEDC) or equivalent focus agency; clarify mandates, empower and allocate adequate staffing and budget	States/cities	ST
Cumbersome licensing, permitting and business processes	Set up city level one stop window for all clearances for setting up businesses; advocacy with central and state agencies regarding business regulations, tax/subsidy related issues	Cities	ST-MT
Lack of business development services	Partner with relevant agencies (SME development agencies, educational institutions, etc.) and private sector to address develop skill development programs, incubation hubs, common facility centers etc. for priority sectors	CEDC/equivalent agency	MT
Access to land and affordable housing			
Cumbersome/outdated land-use regulations and building bylaws	Relax restrictive zoning and development control regulations; streamline and simplify land-use conversion processes	States/cities	ST
Outdated land records and cadaster system	Establish a single land information system (ideally as part of comprehensive urban database); update land cadaster system; identify vacant land that could be repurposed; establish public land inventory.	States/cities	MT
Lengthy land assembly with difficult process of resettlements and compensation	Introduce land-pooling schemes for voluntary land development and ensure effective implementation through building capacity of planning officials, introduction of grievance redressal mechanisms. Enhance creative redevelopment through PPPs.	States/cities	ST-MT

EXECUTIVE SUMMARY

Table ES.1. Summary of key recommendations (*cont.*)

Key issue	Recommendation	Responsible agency	Time frame
Transport			
Lack of legislative backing for transport plans	Align comprehensive mobility and logistics plans with development plans/master plans to enable synchronization of land use and transport planning; invest in high-capacity system on trunk routes and flexible system such as buses for job accessibility.	States/metropolitan areas/cities	ST
Multiplicity of institutions with overlapping functions and weak capacity	Capacitate unified transport authorities in metropolitan areas (like CUMTA in Chennai) or state institutions (like DULT in Karnataka); clarify institutional mandates and strengthen coordination.	States/cities	MT
Limited funding resources and performance management for public transport affect service quality	Provide regular funding to public transport based on targeted levels of service delivery and efficiency (like the Public Transport Service Contract in Chennai); integrate buses and para-transit modes.	States/cities	ST-MT
Infrastructure finance			
Poor implementation and absorptive capacity of available funds	Improve cities' capacity to prepare pipeline investment projects; mainstream performance-based fiscal transfers linked to results; strengthen state-level financial intermediaries to provide technical support to small cities for project preparation, and help them access municipal loans and pooled bonds	Center/states/cities	ST-MT
Lack of staffing and weak financial management capacity undermine investor confidence	Establish a dedicated municipal cadre with technical and accounting staff; standardize fiduciary systems, accounting norms and practices	States	MT-LT
Lack of interest for private capital participation and mobilization	Regular dialogues with financial intermediaries/investors to raise awareness of market potential, government policies, and available revenue streams to support infrastructure financing	States/cities	MT
Urban amenities, livability, and resilience			
Insufficient capacity for climate and disaster preparedness	Update infrastructure design guidelines and building codes to incorporate climate adaptation and mitigation measures; prioritize nature-based solutions or green infrastructure	States/cities	ST-MT
Lack of policies to promote compact development and contain urban sprawl	Put in transit-oriented development (TOD) policy and implementation guidelines to enhance densification, mixed land use, and walkability.	States	ST
Limited access to adequate infrastructure and services	Increase access to reliable water, waste management, sanitation, electricity and digital services; improve access to and quality of education and health services to both increase human capital and attract and retain talent	States/cities	ST-MT

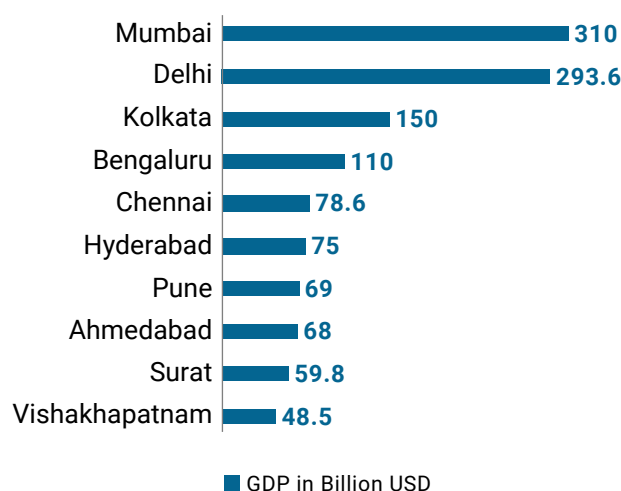
Note: CUMTA = Chennai Unified Metropolitan Transport Authority; DULT = Directorate of Urban and Land Transport; ST = short term; LT = long term; MT = medium term.

I. INTRODUCTION

RATIONALE

India's urban population is growing, and this could be a boon to development.² By 2050, an estimated 50 percent of the population is expected to reside in urban areas, increasing from 37 percent in 2022 (World Bank data, 2022).³ Per estimates, cities contribute 54 percent of India's GDP (2011-12) and 48 percent of employment (MoSPI: Economic Census, 2013). An estimated 30 percent of India's gross domestic product (GDP) comes from 10 cities (figure 1.1).

Figure 1.1. Contribution of cities to India's GDP



Source: www.statista.com.

While historically worldwide, cities have been key drivers of economic growth and job creation, urbanization in India is underleveraged. Cities are “places” where firms and people interact, creating jobs and reaping “place-specific” benefits (World Bank, 2024a). They provide numerous “agglomeration economies”⁴ by bringing firms and workers together, allowing for knowledge spillover effects and improving productivity (Quintero and Roberts, 2018; ADB, 2024). Urban agglomerations allow for deeper labor markets, which facilitate better matching between workers and jobs and the growth of a diversified base of specialized local suppliers that can provide inputs into final production. Compared with rural areas, cities provide better connectivity, access to

critical infrastructure, production inputs, and domestic and international markets, and they generate more and better job opportunities (Gollin, Jedwab, and Vollrath, 2016). This is the case in India; however, in Indian cities, the benefits of agglomeration are not being fully realized due to various “congestion forces” overshadowing benefits of proximity to firms and workers (Ellis and Roberts, 2015; Hasan, Jiang, and Rafols, 2017). These can be driven by urban infrastructure and services gaps, management and regulations that are not fit for purpose, land and property market constraints, and challenges facing city residents in terms of access to affordable housing, urban services, manageable travel times to work, vulnerability to climate-related risks, and access to green and livable public spaces.

For India to reach a \$30 trillion economy by 2047, as envisioned in Viksit Bharat 2047, cities will need to play a fundamental role. Recognizing the importance of cities to India's economic growth, poverty reduction, and jobs, the central government in its FY24-25 budget highlighted “urban growth hubs” as a key priority.⁵ It also established a High-Level Committee (HLC) on Urbanization in the Ministry of Housing and Urban Affairs (MoHUA, 2022) to envision the role of cities as gateways to enable better trade and economic activities. Under Viksit Bharat 2047, which covers multiple areas of development, including economic growth, social advancement, environmental sustainability, and effective governance, a Growth Hubs program has been launched by NITI Ayog to develop urban areas in a structured way to help achieve the economic goal of transforming India into a \$30 trillion economy by 2047. NITI Ayog has already developed a strategy and roadmap to promote selected pilot “city regions” (Mumbai, Surat, Varanasi, and Vizag) as urban growth hubs. So: How can urban India rise to the challenge? How can the central government better support states and cities to enhance their growth potential? Do the approaches to enhancing growth in megacities and mid- to large cities need to be differentiated from those for small cities? How can support to different cities be prioritized? This report builds on the work of the HLC and Niti Ayog and other ongoing efforts and outlines a roadmap for advancing the Viksit Bharat 2047 vision of thriving, economically vibrant urban centers across India.

OBJECTIVE

The main objective of this report is to develop a knowledge framework to support urban growth hubs in India and propose actionable recommendations to implement it. In this study, “urban growth hubs” are defined as cities and urban areas that are competitive, livable, resilient, and inclusive. They have above average economic potential⁶ understood as latent capacity for greater economic development, and key enablers⁷ to realize that potential. Not any human settlement can be an urban growth hub. For cities to be considered a growth hub in this study, they must not only be major contributors to economic growth and/or jobs but also cities where large proportions of firms and workers are engaged in the production of manufacturing and/or tradable services—sectors with considerable potential for raising productivity.

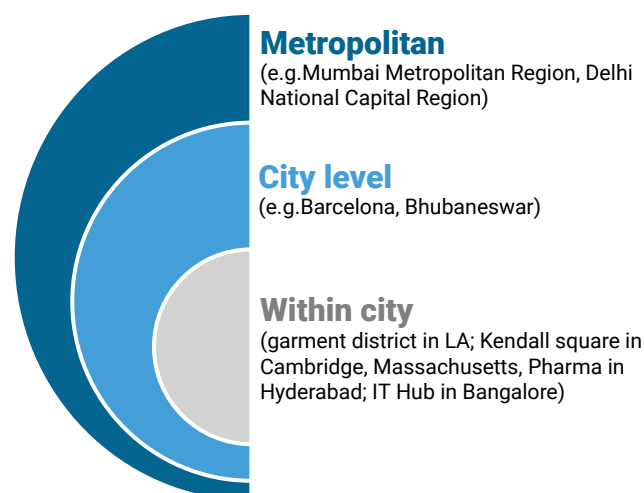
There is no single model of urban growth hubs. They can be driven by manufacturing, services, or transformative infrastructure; or by the upgrading of a key ecosystem such as an urban river. They can occur at multiple scales, such as within the city, e.g., the garment district in Los Angeles or the information technology (IT) hub in Bengaluru; at city level, such as Barcelona; or at metropolitan level, e.g., Mumbai Metropolitan Region (MMR) or the Delhi National Capital Region (DNCR) (figure 1.2). In some cases, greenfield cities can also be growth hubs (annex 1).

AUDIENCE AND ORGANIZATION

This report is intended for two main readerships. At national level, it will inform programs and schemes led by MoHUA in collaboration with other ministries and agencies to promote urban growth hubs in India; at subnational level it will inform states and cities on issues and criteria to consider in selecting and scaling up urban growth hubs.

The rest of this report is structured as follows. Chapter 2 highlights key features of India’s city systems and the urban growth hub framework and typologies. Chapter 3 discusses key cross-cutting issues that cities must address to better leverage their agglomeration benefits. Chapter 4 provides a template and step-by-step guidance to states/cities on how to develop an urban growth hub strategy. Chapter 5 outlines principles for developing a national scheme on urban growth hubs. Chapter 6 concludes with a summary roadmap and recommendations.

Figure 1.2. Growth hubs can occur at different scales

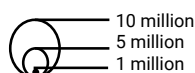
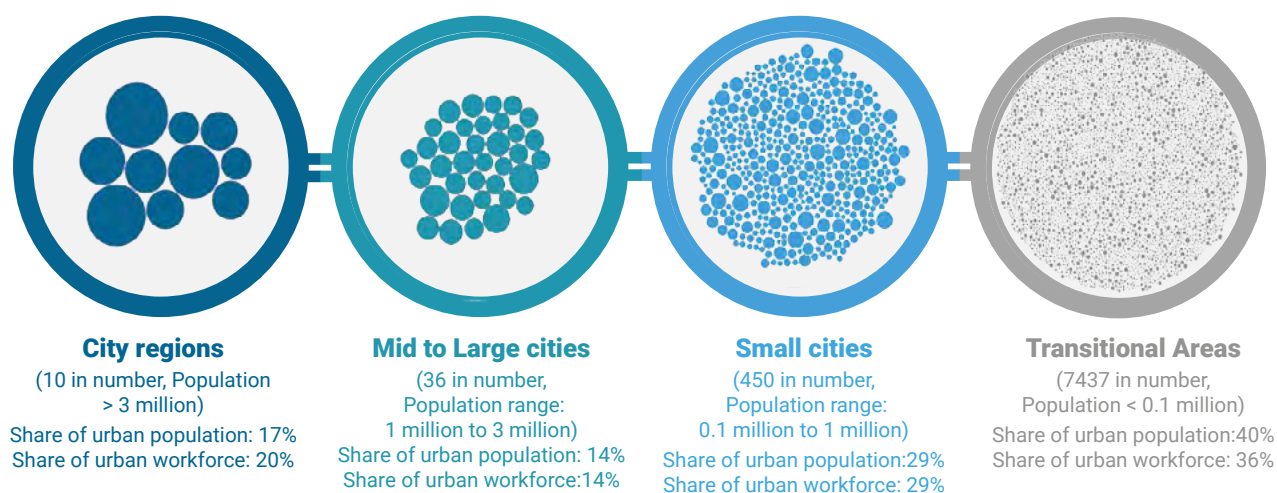


II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

India has one of the largest urban systems globally. As per the Census of India (2011) there were an estimated 4,041 statutory towns and 3,892 census towns⁸ (figure 2.1) in 2011 and these numbers have since increased.⁹ The Census (2011) classifies towns into six tiers based on population size where tier 1 refers to all cities above 1 lakh (i.e., 100,000) population, thereby not distinguishing between the growth characteristics of mid- to large and smaller cities. Taken together, over 65% of India's urban population is concentrated in small cities, small towns and census towns (ibid.) less productive individually but numerous enough to have a significant growth impact at the national level and increasingly important in terms of employment generation (IIHS, 2015). An up-to-date nationwide database on Indian cities is not available and needs to be developed and regularly updated for a more accurate characterization of cities, urbanization, and growth patterns in India.

Based on the 2011 Census, publicly available data and administrative boundaries at the time, statutory towns can be categorized in a more differentiated way for analyzing growth trends and patterns (table 2.1). Due to changes in population and urban built-up area expansion since, the urban population of the contiguous built urban area would now be higher. Absent city-level GDP data, city classifications are described in terms of population, size of working population, and other factors such as number and size of firms (manufacturing and services).¹⁰ Given the paucity of data, especially for small cities, statutory towns, and census towns, these should be read as *descriptors* and not as *filters* to strictly distinguish between categories.¹¹

Figure 2.1. Statutory and Census towns



Source: Census (2011).

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

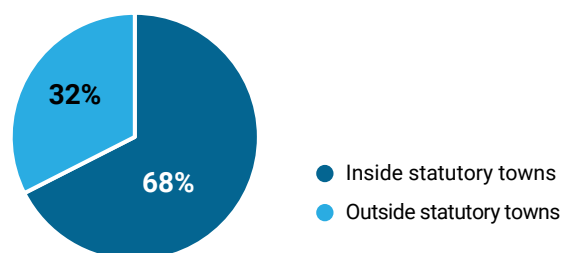
Table 2.1. City classification and their characteristics

Size class	Population in lakh and million (2011 Census)	No. of cities	Working population	Firm number and size (manufacturing)	Firm number and size (services)	Contribution to GDP	Governance structure	Examples
Megacities (Class 1 per 2011 Census)	30 lakh and above (3 million and above) (17% of urban population)	10	> 9.5 lakh working population	> 50 firms with 100–500 people	> 75 firms with 100–500 people	Top 10 contributors to national GDP	Regional authority/ dev. authority/ municipal corporation	Mumbai, Delhi, Kolkata, Chennai, Hyderabad, Bengaluru, Jaipur, Ahmedabad
Mid- to large cities (Class 1 per 2011 Census)	10–30 lakh (1–3 million) (14% of urban population)	36	9.5–3.0 lakh working population	> 50 firms with 10–100 people AND	> 300 firms with 10–100 people	Among top 5 contributors to state domestic product	Dev. authority/ municipal corporation	Coimbatore, Indore, Patna, Varanasi, Visakhapatnam, Kanpur, Kota, Ludhiana
Small cities (Class 1 per 2011 Census), includes 20 census towns with population 1–6.5 lakh	1–10 lakh (0.1–1 million) (29% of urban population)	430 + 20	3 lakh–50,000 working population	> 50 firms with 10–100 people EITHER/ OR	> 100 firms with 10–100 people	Among top 5 contributors to district domestic product	Municipal corporation/ municipal council	Tiruppur, Cuttack, Mangalore, Gaya, Agartala
Transitional areas (Small towns and census towns)	Below 1 lakh (Below 100,000) (small towns and census towns 26% and 13% of urban population)	3,565 + 3,872	50,000–2,500 working population	Data not available for census towns	Data not available for census towns	Local contributor to district domestic product	Municipal corp./ municipal council/ rural local body	Doda; Farooqnagar (census towns); Amalner
Outgrowths	1% of urban population							
	Total	7,933						

Source: Based on 2011 Census data, URDPFI guidelines, 2015, WRI India research paper (forthcoming).

Satellite-based analysis shows evidence of widespread urban sprawl across city sizes and in 2020, an estimated 32 percent of the built-up area was beyond the statutory boundaries of urban local bodies (figure 2.2). These areas are unplanned and under-served due to insufficient governance, technical, and financial capacities of rural local bodies. They are characterized by inadequacies in infrastructure services such as water supply, sewerage and sanitation facilities, solid waste management, and low road densities, and are prone to environmental degradation.

Figure 2.2. Share of built-up area within and outside statutory boundaries of cities in the contiguous built urban agglomeration

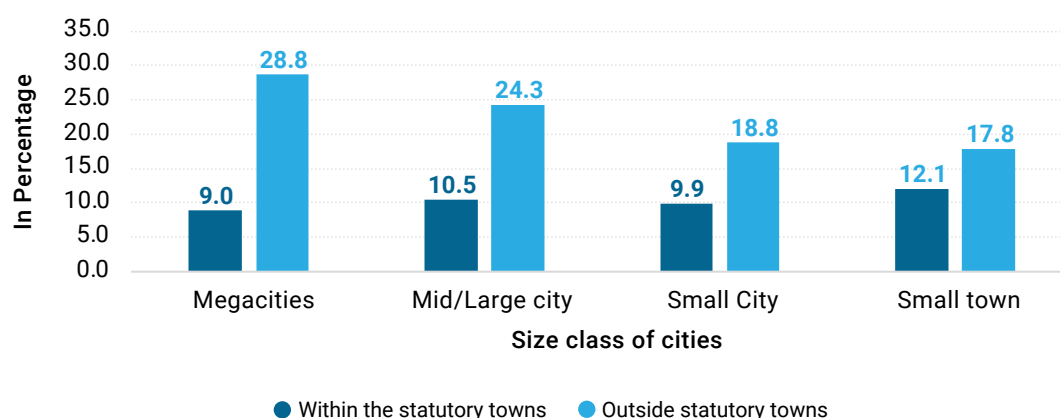


Source: Team analysis, 2025.

The contiguously built urban agglomeration of the megacities—Mumbai, Delhi, Bengaluru, Kolkata, Chennai, Hyderabad, Pune, and Ahmedabad—which are the country’s economic powerhouses, experienced the highest increase

in built-up area outside city limits. Satellite-based analysis shows that the agglomeration effects of the megacities were higher than for the other city categories in 2010-2020 (figure 2.3).

Figure 2.3. Decadal increase in built-up areas, 2010-2020



Source: WRI India, forthcoming.

The agglomeration characteristics of India’s megacities, and mid- to large cities, highlight the need for integrated governance and growth models for these city classifications. India’s megacities (e.g., Chennai, Mumbai, and Hyderabad) and mid- to large cities (e.g., Indore, Patna, Coimbatore, and Bhubaneswar) incorporate multi-jurisdictional agglomerations functioning as an interconnected economy and labor pool. The case of Kolkata and Coimbatore contiguous built urban agglomeration (CBUA) shows that the built-up area for both cities spreads across multiple jurisdictions and is governed by different urban and local bodies; is subject to

different legislation and regulatory frameworks; and incorporates complex coordination, governance, planning, and administration needs (tables 2.2 and 2.3; maps 2.1 and 2.2). For such agglomerations, planning mainly for the 2011 Census-based municipal administrative area based on the 1960 Town and Country Planning Law,¹² which is currently the practice, is insufficient. As proposed by several committees and reports (HLC, 2023; Niti Ayog, 2023), it is recommended that a “city region” based on integrated spatial, economic, and transport planning of the urban agglomeration is put in place.

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

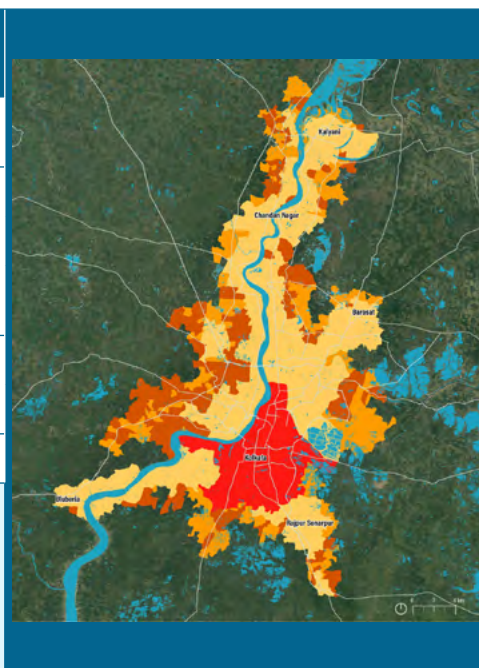
Table 2.2. Population distribution across multiple entities in Kolkata contiguous built urban agglomeration

Sl. No	Type of entity	No. of entities	Population	Share of population	Built-up area (sq km)	Share of built-up area
1	Municipal corporation	3	57,40,636	37%	91.16	34%
2	Municipality	39	72,99,263	47%	126.48	47%
3	Cantonment	1	17,380	0.11%		
4	Industrial township	1	1095	0.01%		
5	Census town	152	18,37,183	12%	51.72	19%
6	Village	227	7,33,484	5%		
	Total	423	1,56,29,041	100%	269.36	100%

Legend

- Kolkata M. Corp
- Contiguous Statutory Towns
- Census Town
- Villages
- Water Bodies
- Major Road Networks

Map 2.1. Multiple jurisdictions in Kolkata contiguous built urban agglomeration



Source: WRI India, forthcoming.

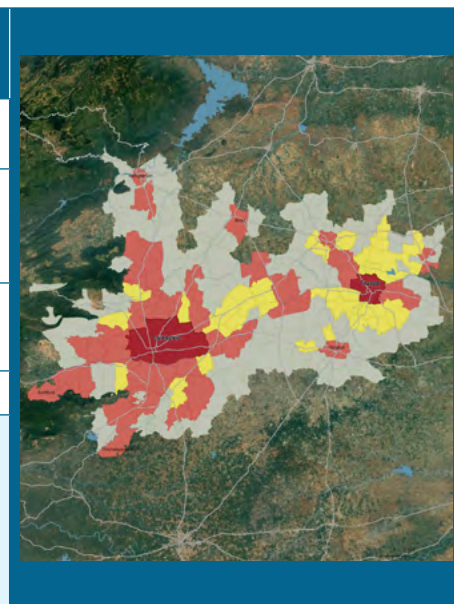
Table 2.3. Population distribution across multiple entities in Coimbatore contiguous built urban agglomeration

Sl. No	Type of entity	No. of entities	Population	Share of population	Built-up area (sq km)	Share of built-up area
1	Municipal corporation	2	14,95,073	38%	42.8	22%
2	Municipality	7	5,72,479	14%	73.96	38%
3	Town panchayat	38	7,68,059	19%		
4	Census town	34	5,84,755	15%	79.34	40%
5	Village	105	5,40,714	14%		
	Total	186	39,61,080	100%	196.1	100%

Legend

- Coimbatore M. Corp
- Other statutory towns
- Census Town
- Villages
- Water Bodies
- Major Road Networks

Map 2.2. Multiple jurisdictions in Coimbatore contiguous built urban agglomeration



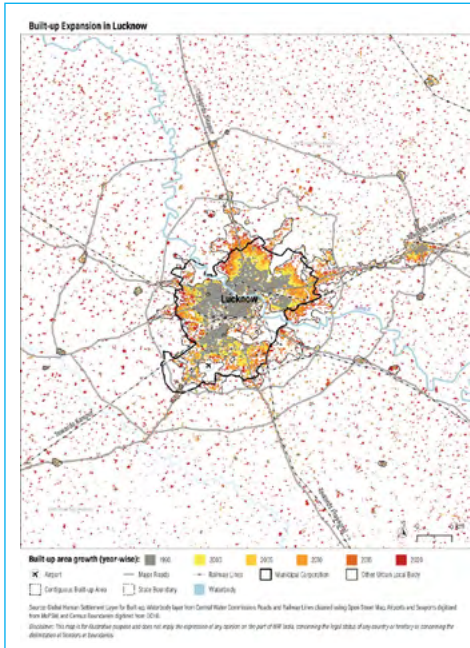
Source: WRI India, forthcoming.

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

Jobs are closely correlated with growth in population and built-up area growth, and larger cities and their immediate peripheries are the preferred destination for migrants. In Uttar Pradesh for instance, the districts of Lucknow, Kanpur,

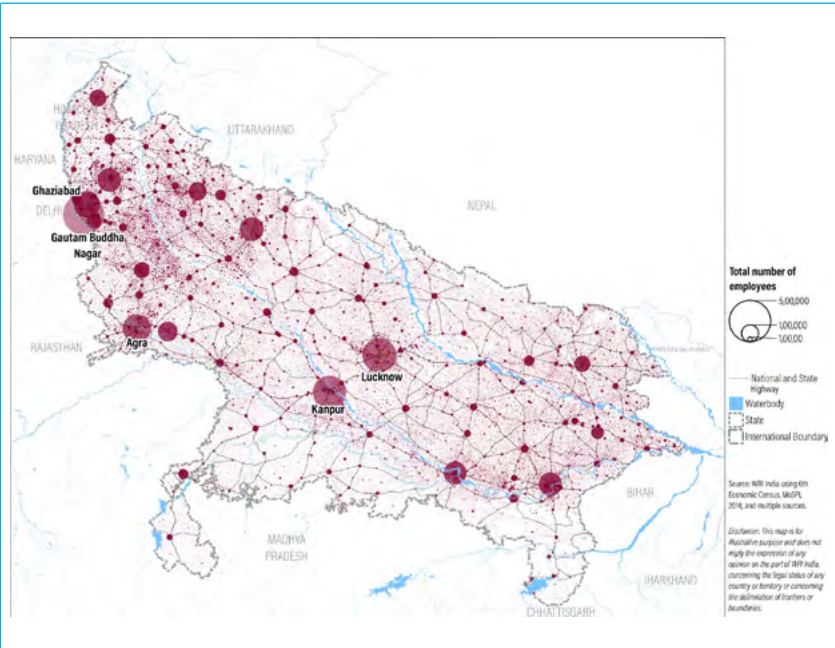
and Gautam Buddh Nagar are experiencing some of the highest expansions of built-up area. These are also the main employment nodes and destinations for in-migration in the state (maps 2.3 and 2.4).

Map 2.3. Peripheral built-up area expansion in Lucknow



Source: Team analysis based on Global human settlement layer, Central Water Commission, Open Street map, Ministry of Ports, ships and waterways, Census 2011.

Map 2.4. Distribution of employment in Uttar Pradesh

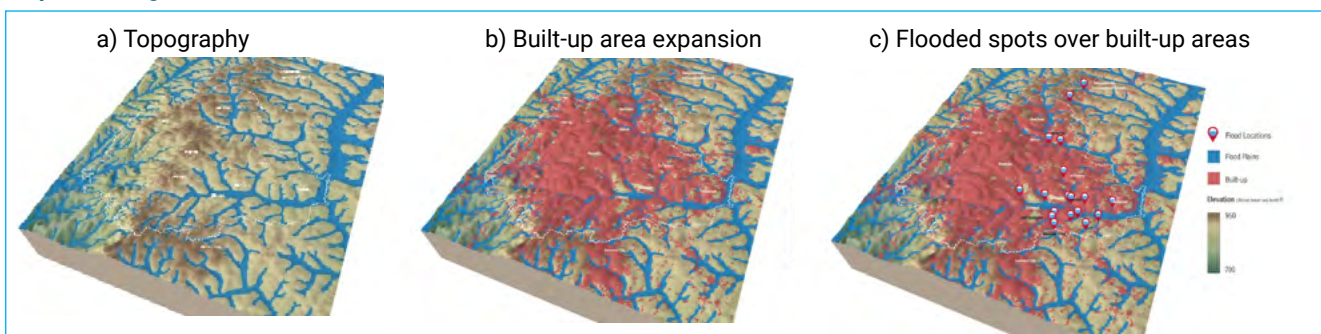


Source: Team analysis based on Sixth Economic Census 2013-14.

Large infrastructure investments such as industrial corridors and highways have been catalytic in driving rapid growth in urban built-up area, but have also compromised environmental assets in cities, contributing to climate vulnerability. Bengaluru, for instance, is situated at the top of a ridge and its topography has a series of lakes, valleys, and

ridges. The city's built-up area has expanded into its valleys and drains, compromising the natural flow and infiltration of water into the soil. This contributes to hotspots where urban floods occur during high precipitation events (map 2.5). Planning for urban growth should thus consider climate risks and vulnerability.

Map 2.5. Bengaluru



Source: WRI India analysis, 2022.

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

OUTLINE OF THE URBAN GROWTH HUB FRAMEWORK

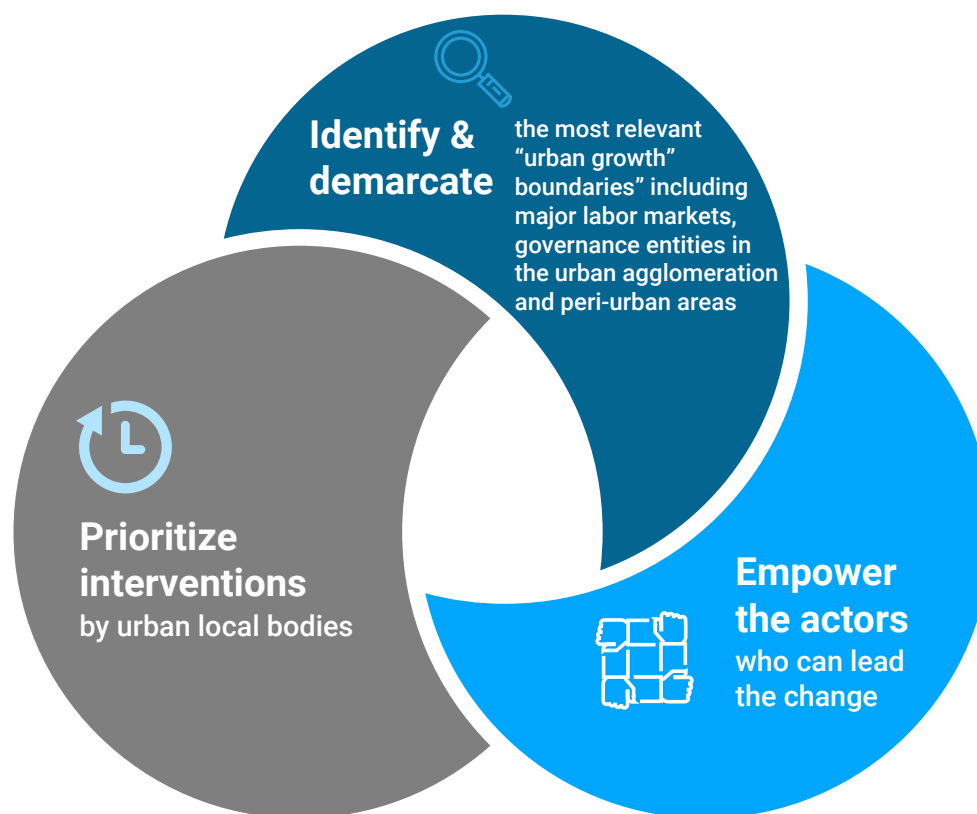
Cities in India, as elsewhere, are characterized by differences in untapped agglomeration benefits and excessive congestion forces. Cities are integrated labor markets. However, due to extensive unplanned development of urban peripheries across city size classes, the geographic boundaries of local labor markets do not necessarily coincide with those of existing administrative units as maps 2.1 and 2.2 indicate. Moreover, the scope for growth varies across urban areas given variations in governance structures, municipal capacities, access to markets, economic diversity, connectivity, skills and labor pool. Even though megacities, mid-to large and smaller cities face many similar challenges such as peri-urban growth, there are differences between these cities in terms of scale, governance structures, and growth drivers. Megacities, for instance, have better market access and connectivity, better strategic infrastructure, a larger labor pool, and greater diversity of industries, contributing to

greater innovation, higher productivity, and more job opportunities than smaller cities. Cities also face many cross-cutting constraints (as discussed in the next Chapter) and hence need to prioritize potential interventions.

The proposed framework (figure 2.4) suggests that city administrations will need to take the following key actions:

- *Identify and demarcate* the most relevant “urban growth boundaries” and plan for it in an integrated way to better address challenges relating to peri-urban growth—a key concern raised in the FY 24-25 budget statement.
- *Engage and empower relevant institutional agencies and actors* who can lead this change.
- *Prioritize interventions* and investments with a focus on the relevant spatial boundaries given limited resources and capacity at most urban local bodies.

Figure 2.4. Key pillars of the urban growth hub knowledge framework

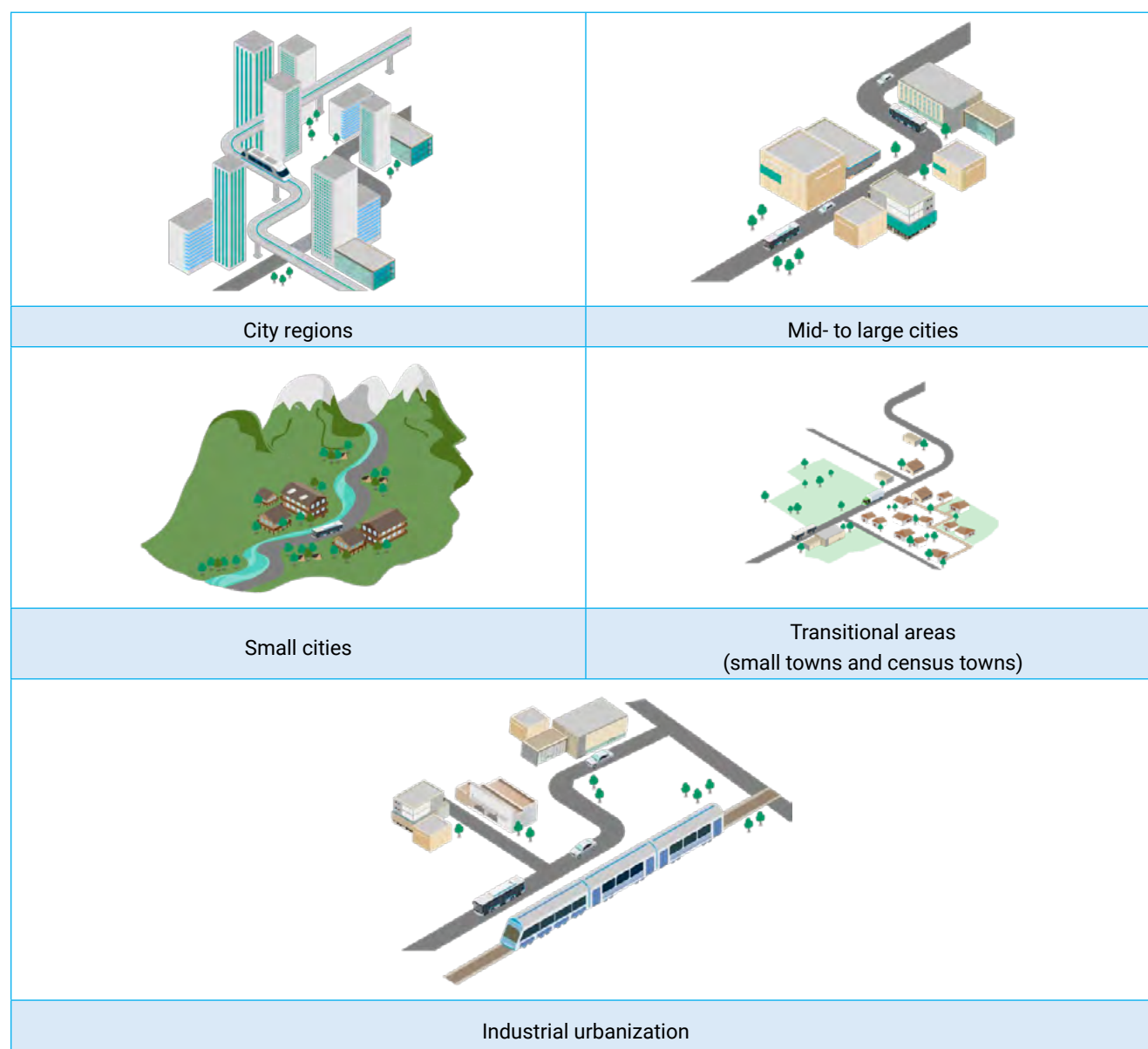


Source: Team Analysis, 2025.

National and international experience shows that there is no “one-size-fits-all” approach to urban growth hubs. A case-by-case approach will also not work for India’s diverse urban system. Rather, a typology of cities is needed to take the strategy to scale. Based on India’s city system, five different categories of urban areas are proposed and briefly explained below (figure 2.5). These include: “city regions”

(HLC, 2023; Niti Ayog, 2023); “mid- to large cities”; “small cities”; “transitional areas,” which include small towns and census towns aligned with MoHUA’s existing city classification;¹³ and “industrial urbanization.” While transitional areas are not growth hubs, special programs can be put in place to support them.

Figure 2.4. Typologies of urban areas



Source: Team Analysis, 2025.

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

With this approach to incentivizing urban growth hubs, cities can devise their own growth pathways and strategies tailored to their urban context. Collaboration between cities within each typology over time, can contribute to learning and innovation between cities, and identify common strategies and growth pathways for specific typologies. Each will need to devise tailored growth strategies.

While at national level, a small town may not be considered a growth hub, in some states and union territories, a small town could be the largest urban area within the state/union territory and have the potential to drive growth locally, and so small towns are retained as a growth hub typology. For instance, Aizawl in Mizoram and Agartala in Tripura, despite being relatively small in the national context, are the largest urban centers within their states, have the potential to drive significant regional and local economic growth, and can therefore be considered potential growth hubs. Small towns like Anand in Gujarat and Darjeeling in West Bengal provide important market and service linkages to promote the development of the hinterland (Scrase et al., 2015). With targeted investments and strategic planning, these towns can boost the economies of their regions.

These typologies are not mutually exclusive. They are intended to aid decision-makers in prioritizing and targeting interventions. A brief description of different typologies is provided below. Detailed city-level diagnostics must be carried out as follow-up to identify growth pathways for specific cities.

CITY REGIONS

City regions include India's 10 largest contiguous built-up urban areas (see table 2.1) including Delhi, Mumbai, Bengaluru, Hyderabad, Chennai, Kolkata, Ahmedabad, Pune, Surat, and Jaipur. Megacities such as the Delhi National Capital Region (DNCR) contributed 8 percent of GDP in 2016 (NCRPB Plan) and the Mumbai Metropolitan Region (MMR) contributed 5 percent (Niti Ayog, MMR Growth Hub report). While all 10 cities had population greater than 3 million people, Delhi and Mumbai had population over 10 million within the municipal administrative boundary (as per 2011 census).

These cities have a highly diversified economic structure, are supported by a large concentration of firms across a range of sectors including manufacturing, tradable services—IT, IT enterprise solutions (ITeS), and financial services—and non-tradable services, and a skilled workforce. They are governed by multi-municipal and multi-jurisdictional planning

frameworks and governance structures (see table 2.1). Cities in this group benefit from access to major connective infrastructure such as international airports, rail, roads, and ports. Greater market access, economic density, and complexity give city regions a substantial economic advantage. However, various congestion forces, regulations, and practices undermine densification and co-location. Their built-up area grew by more than 1.8 percent a year between 1990 and 2020 (a full 3.02 percent in Bengaluru's case with some de-densification in core areas arising from high land prices and difficulty in land-use changes in the city's core).¹⁴

The ongoing development of large metro networks in many such city regions provides an opportunity to rethink the city plans around each brownfield station, based on their connectivity, location, and market potential¹⁵ and to foster the development of well-planned and well-connected new towns around greenfield stations.

MID- TO LARGE CITIES

Mid- to large cities with a population of 1 million–3 million (per the Census 2011 municipal administration boundary) share many of the characteristics of megacities. There are 36 such cities (see table 2.1). They are among the top five contributors to state domestic product and have a working population of 0.3 million–0.9 million. Their economies are less diversified than those of megacities (see table 2.1), but they share many similar constraints with the megacities such as unplanned peri-urban expansion, inadequate land administration, limited infrastructure and urban services, urban congestion, and governance challenges. At the agglomeration level, a mid-sized city such as Indore also includes various small towns and census towns, highlighting the need for integrated economic, urban, and transit planning, and for cities of different sizes within an agglomeration to work well together.

The emergence of greenfield satellite cities around megacities such as Mumbai, Delhi, Bengaluru, and Hyderabad as well as mid-sized cities such as Indore and Bhubaneswar is a growing feature that needs to be considered as part of metropolitan area planning, with close attention to brownfield development within cities and connectivity issues between the municipal core and peripheries, including satellite cities.

SMALL CITIES

Small cities have a population range of 0.1 million–1 million. There are 430 such cities (450 if 20 census towns which meet population criteria are also included) and are distinguished

from other city categories based on working population and firm size (see table 2.1). In some states, they are among the top five contributors to district domestic product. Examples of small cities, many of which have distinct economic potential, include Panipat (Haryana) for textile and handloom weaving, Gaya (Bihar) for religious tourism, Shimla (Himachal Pradesh) for nature tourism, Dehradun (Uttarakhand) for tourism, and Dhanbad (Jharkhand) for mining. They are typically governed by a municipal corporation or council. In small cities, about 20-30 percent of their employment is concentrated in a few sectors such as tourism and specialized products.

Small cities are focused on a few distinct sectors driving employment and output, with lower levels of economic diversity, but have the potential to contribute significantly to the regional economy. Beyond the focus sectors, they are prominent in retail trade, hotels, and the food and beverages manufacturing and services sectors. Their economies are often centered on built heritage or nature tourism, religion and culture, natural endowments, and even education. The cities are usually less populated with limited infrastructure and less connectivity to major cities. They have shown growth in their built-up area over the last three decades, with some cities such as Dehradun and Shimla growing at 2.4 percent a year.

TRANSITIONAL AREAS (SMALL TOWNS AND CENSUS TOWNS)

Transitional areas cover small towns, both statutory towns (3,565) and census towns (3872) with a population below

100,000 as per 2011 Census. Key features include a working population of more than 2,500, and more than 50 firms of any size and type (manufacturing or services). Statutory towns are governed by some form of urban local body (ULB) while census towns are governed by rural local bodies.

Transitional areas (MoHUA, 2015)¹⁶ are connected to their rural hinterlands and contribute to the local economy. They are in transition from a rural to a more urban economy and include small towns that are not in the immediate periphery of larger cities. The economy is dominated by agricultural and handmade products. They serve as permanent and seasonal markets for nearby rural produce, provide labor to nearby industries, and tend to have a workforce skilled in agrarian and agro-processing activities and basic trade. Examples include Gangtok and Rangpo in Sikkim and Baddi in Himachal Pradesh.

INDUSTRIAL URBANIZATION

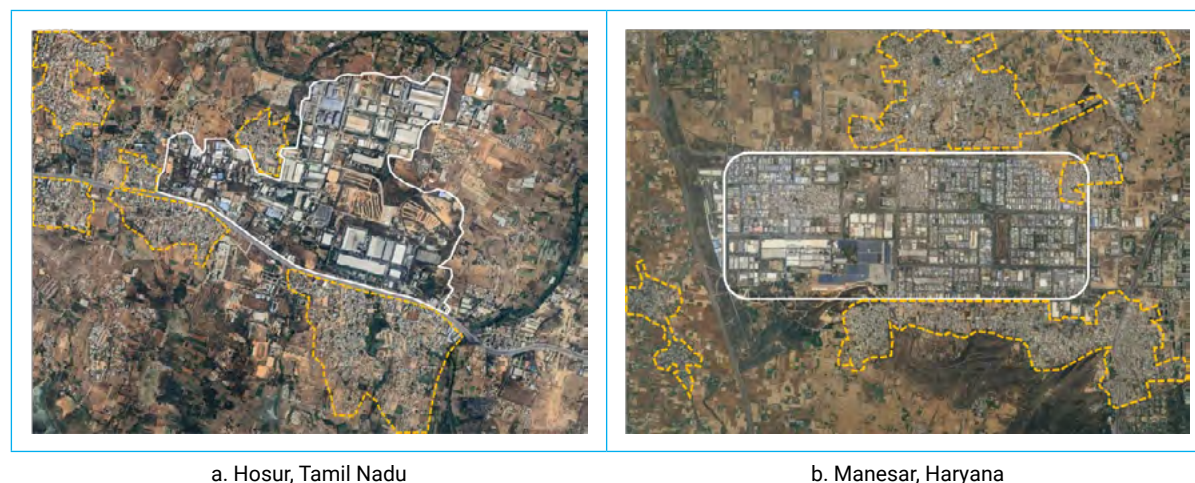
India's industrial areas provide untapped potential for driving urban economic growth. These include cities/towns around special economic zones (SEZs); proposed towns and cities located along the 11 major industrial corridors planned under the National Industrial Corridor Development Program (NICDP); and urban—mainly unplanned—outgrowths around industrial estates (table 2.4, images 2.1 and 2.2).

Table 2.4. Types of industrial urbanization

Type	Number	Key features	Governance	Examples
SEZs	About 370	SEZ Policy 2005 predates the NICDP corridor initiative	Ministry of Commerce and Industry/ State Industrial Development Boards	NOIDA SEZ (UP); SEEPZ SEZ (MH) Manikachan SEZ (WB)
Ind. towns/ nodes	No accurate data	» Access to multimodal connectivity such as NICDC alignments, and regional connectivity via air/ water	Under governance of State Industrial Development Board	Hosur (TN); Manesar (HR);
NICDP corridor	32	» Resource proximity (to minerals, coast, solar, etc.) » Decongesting an existing city or leveraging proximity to an economic hub » Potential to attract anchor investments by the private sector (national, international)	Ministry of Commerce and Industry/ State Industrial Development Boards/ Special Investment Region Development Authority	DMIC Corridor: a) Dholera Special Investment Region (GJ) b) Shendra Bidkin (MH)
NICDP = National Industrial Corridor Development Program.				

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

Image 2.1. Unplanned industrial urbanization



Source: Google Earth.

Image 2.2. Shendra-Bidkin Industrial Area-Aurangabad



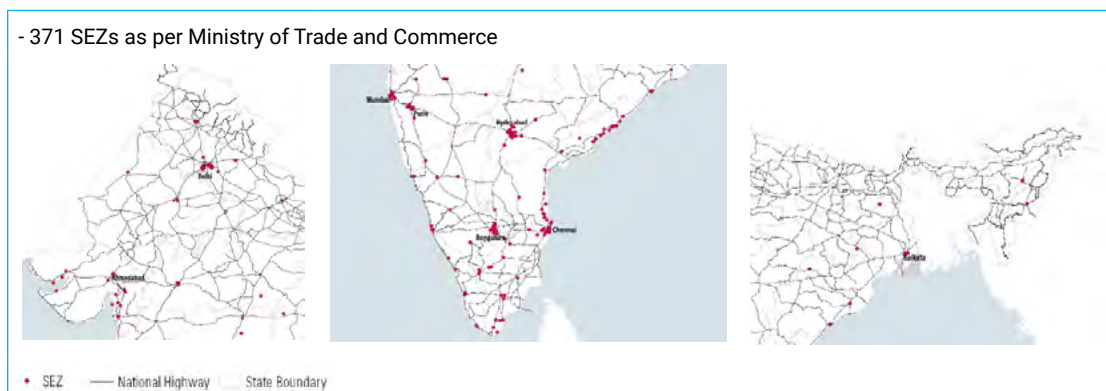
Source: NICDC.in.

The central government has undertaken numerous initiatives to drive industrial development. Setting up Industrial Area Development Boards (IADBs) were an effort at industrial cluster development and orderly economic and infrastructure development in the states. In 2000, the central government launched an SEZ policy to drive economic growth, supported by high-quality infrastructure and an attractive fiscal package at both central and state levels, with minimal regulation. Studies show that municipalities with SEZs had 52 percent higher employment growth in manufacturing and services than municipalities without SEZs (Galle et al., 2022).¹⁷ In 2007, with the NICDP, the government focused on growth of industrial cities through the development of industrial corridors that can compete with the best manufacturing and investment locations in the world. Expanding the focus from industrial clusters as nodes to large economic growth region, the government in 2009 passed the Special

Investment Regions (SIR) Act,¹⁸ to establish large-scale Investment Regions/Industrial Areas and develop them into global economic hubs with world-class infrastructure.

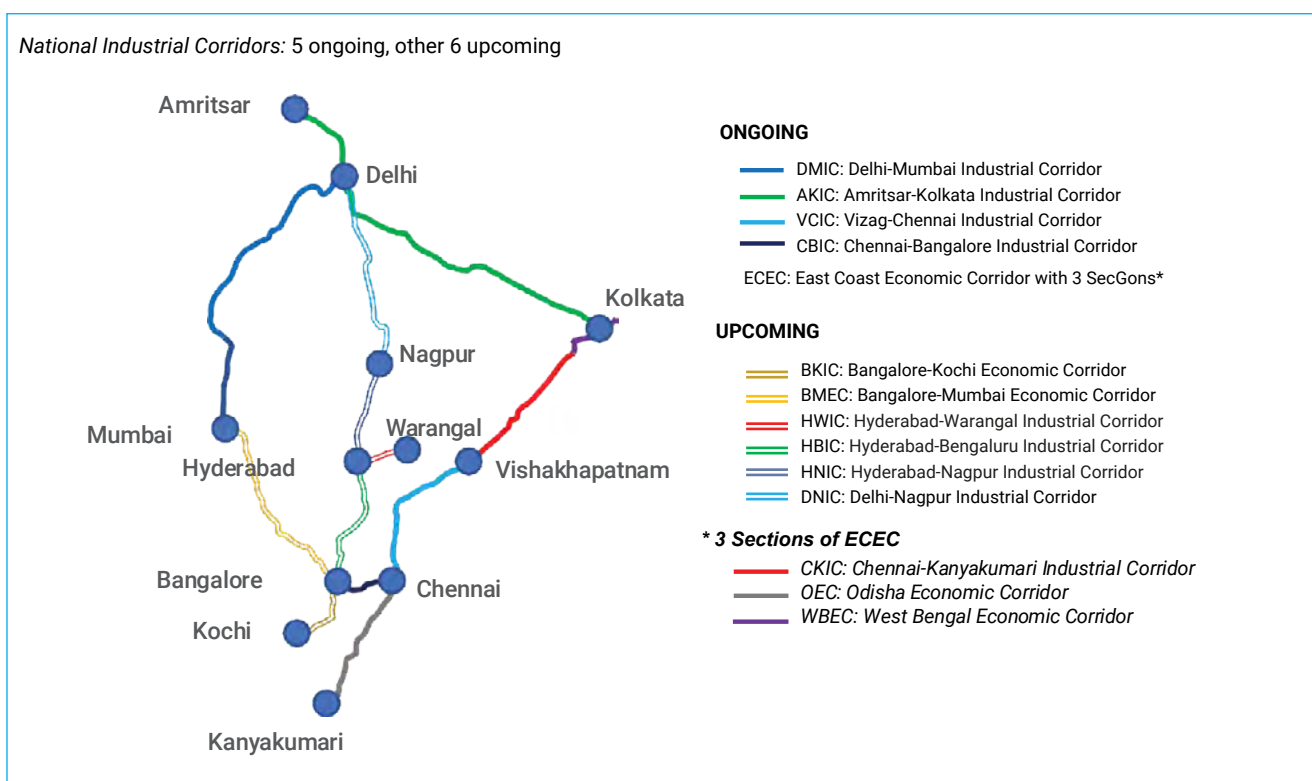
India's system of cities is key to achieving its ambitious industrial and economic development goals. Though these industrial layouts incorporate orderly development within the cluster, in many cases, there is unplanned urban development around them (map 2.6). Well-located and well-designed industrial areas/corridors can serve as a catalyst for jobs. If well planned, the surrounding area could be transformed into livable cities and growth hubs. States/cities should seize this opportunity to systematically plan and develop urban areas around industrial estates, nodes and corridors for leveraging the synergies between industrial and urban development in India.

Map 2.6a. SEZs across all states



Source: Based on Dept of Commerce and MTC 2022 data.

Map 2.6b. National industrial corridors



Source: NICDC.in.

DIFFERENCES IN GROWTH PATHWAYS: BENGALURU, BHOPAL, AND BHIWANDI

City regions, mid- to large cities, and small cities may have similar characteristics such as peri-urban growth, economic diversity, city level amenities, congestion, waste-management and capacity and capability gaps, but there are also

clear scale differences between them. This is illustrated through a comparison of the cities of Bengaluru, Bhopal, and Bhiwandi.

The megacity of Bengaluru had a population of 8.44 million (2011 Census), has an international airport, city metro rail

II. GROWTH HUB FRAMEWORK AND TYPOLOGIES

networks, a planetarium, museums, convention and exhibition centers, multi-specialty hospitals, the highest density of colleges in the country—1,106 per the All-India Survey on Higher Education (AISHE)—and a highly skilled workforce. Industrial sectors range from IT and IT enterprise solutions, biotechnology, aerospace, telecommunications, electronics, garment manufacturing, automobiles, food processing, banking and finance, mining, steel, and cement production. Bengaluru has a regional development authority, a development authority, and a municipal corporation.

The mid-sized city of Bhopal had a population of 1.79 million (2011 Census), has a domestic airport, is setting up a limited city metro rail network, has museums, a mini planetarium, hospitals, and a reasonable number of colleges (327 per AISHE), and a skilled workforce. Its economy is diversified in sectors such as electrical machinery, transport and business equipment, electrical goods, jewelry, chemicals, and cotton. Bhopal has a development authority and a municipal corporation.

The small city of Bhiwandi, Nizampur had a population of 0.7 million (2011 Census), relies on the nearest megacity of Mumbai for access to an airport, and other such higher-order facilities. It functions economically as a hub for textile

and garment manufacturing, has become a destination for mills that were relocating out of the megacity of Mumbai, and has a semiskilled workforce. Bhiwandi has only a municipal corporation but falls within the region of Mumbai and so is overseen by Mumbai's Regional Development Authority (MMRDA). Bengaluru has an HHI¹⁹ of 0.08 (highly diverse), Bhopal has an HHI of 0.14 (less diverse), and Bhiwandi has an HHI of 0.20 (least diverse).

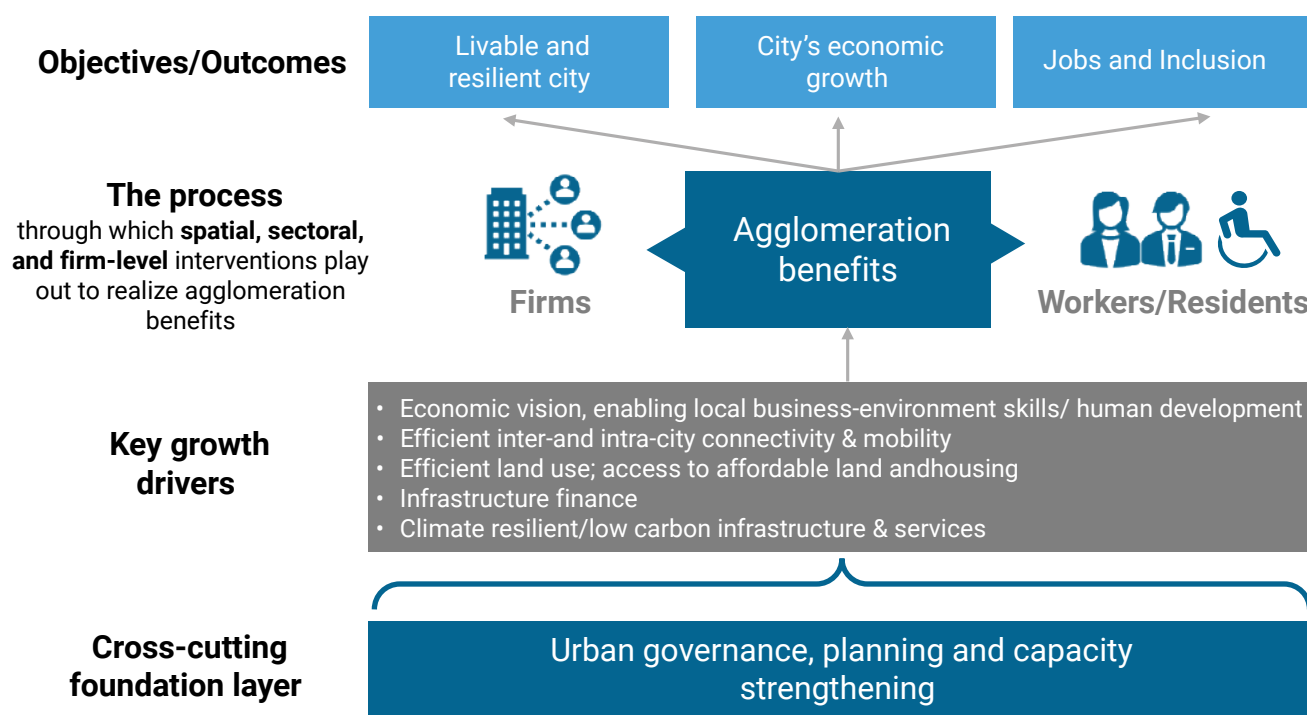
While each of these three cities has the potential to become a growth hub in their respective regions, their growth pathways will, however, differ. This will be in terms of how far up the value chain they can move with their products, the types of amenities and services they can sustain, the nature of investments they can attract such as in specialized semiconductors or just basic metal production, and thus the levels of skilled talent and knowledge workers that they can attract. Productivity gains are, for example, seen to be higher in cities with high-tech sectors and services industries than in other economies with a larger or more mature manufacturing industrial base. Economic growth depends on context, and factors such as urban infrastructure, institutional capacity, and industry composition (Frick and Rodríguez-Pose, 2018).

III. KEY ENABLERS FOR REALIZING CITY ECONOMIC POTENTIAL

International experience suggests that a key factor for urban growth hubs to flourish is exhibiting economic potential. Five typical contributors to economic potential include market access, urban population, economic density, transport connectivity, and human capital (Roberts, 2016). Even where there is potential, there are often constraints to growth that may require government interventions to relax them. Figure 3.1 offers a lens for identifying constraints to the key contributors for achieving agglomeration benefits and city

economic development objectives. Specifically, it identifies urban governance, planning, and institutional capacity strengthening as a foundational cross-cutting issue and five contributing factors or “building blocks” for achieving agglomeration benefits and a city’s overall goal of economic growth, jobs, resilience, and livability. This chapter describes these factors and proposes recommendations—also summarized at the end of the report—for catalyzing urban growth in the context of Indian cities.

Figure 3.1. Building blocks for realizing city economic potential



Source: Team analysis based on World Bank 2024a and ADB 2024.

STRENGTHENING URBAN GOVERNANCE, PLANNING, AND INSTITUTIONAL CAPACITY

Effective urban governance and planning are critical to realize the economic growth potential of cities. In India, systemic capacity constraints prevail at state and urban local

body levels impacting urban management. In some cases, state urban departments continue to undertake planning for small and medium sized cities, and the transfer of urban planning functions as envisioned in the 74th Amendment to the Constitution has not fully materialized (Niti Ayog, 2021). Despite the importance of master plans as statutory

III. KEY ENABLERS FOR REALIZING CITY ECONOMIC POTENTIAL

instruments to guide spatial and economic development of a city, 65 percent of urban settlements do not have any master plans, resulting in fragmented urban development, and exacerbated congestion forces and sprawl (ibid.). Most census towns—though recognized as “urban” in the 2011 Census—are governed by rural bodies that face many capacity constraints (Niti Ayog, 2021; Janagraha, 2023).

The 1960 model Town and Country Planning Law (amended in 1985) and associated bylaws which guide master planning, should be updated. This updating would address India’s current urbanization patterns and incorporate integrated planning and sustainability. States’ town and country legislation should also be updated as needed, to ensure that peri-urban areas are planned in an integrated way, along with the municipal administrative boundaries (Niti Ayog, 2021; HLC Report 1, 2023). Odisha’s rural–urban transition policy—a good example of a proactive effort to systematically plan peri-urban areas—could be adapted and scaled up to other states.²⁰

Standardization of a tiered spatial planning approach will support urban livability and the economic growth agenda in a city (figure 3.2). At present, there is considerable variation in urban planning and underlying legal frameworks across states. As firms cluster and families settle in cities, state and local governments have an essential role to ensure that land supply is adequate, that infrastructure is provided for connectivity and mobility, and that services are made available for businesses and residents. Spatial planning is a critical instrument enabling city authorities to get one step ahead of unplanned urbanization, to guide the sustainable supply of serviced land, and to facilitate locational choices of businesses and families. Cities currently prepare master plans, land use, and/or development plans focused on statutory town boundaries identified in the 2011 Census. These plans typically do not include metropolitan-level plans that emphasize integrated economic, urban, and transit planning including peri-urban areas, nor are systematically complimented by micro-level or area-based plans (HLC Report 1, 2023). To address the needs of rapidly sprawling cities, “city-regions” should have a three-tiered planning system: (1) larger scale integrated urban/spatial, transport and economic planning for the metropolitan area (including peri-urban areas) that defines growth areas (expansion and densification) and the arterial infrastructure, thereby guiding investment at adequate locations; (2) master plans for the cities per the administrative boundaries; and (3) various micro-level plans

that facilitate local implementation aligned with each other (HLC Report 1, 2023).²¹ For small cities and towns, a two-tiered planning system including a master plan for the city per the administrative boundaries and various micro-level plans should be put in place (figure 3.2).

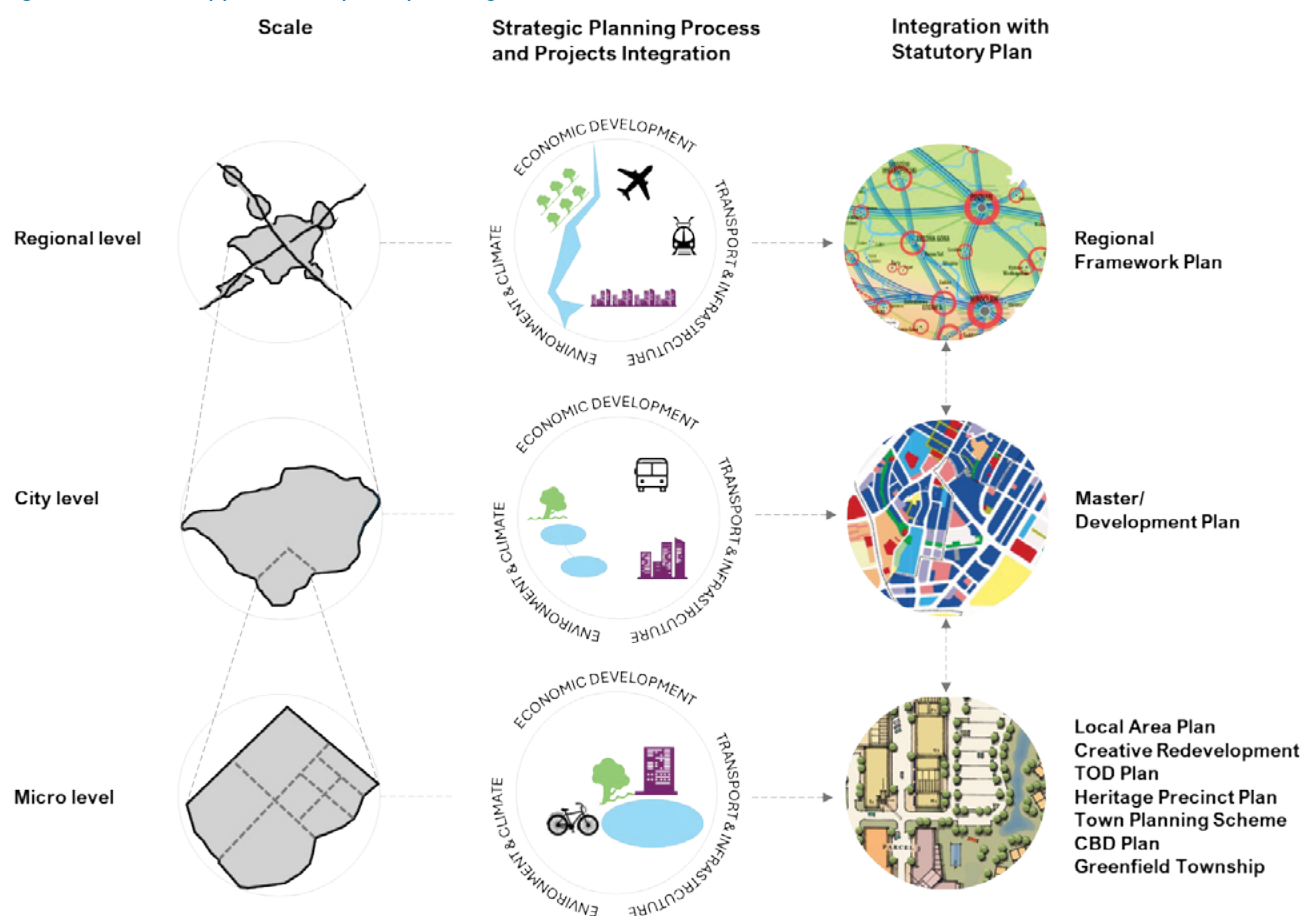
A designated entity for integrated urban, transport and economic planning that includes peri-urban areas, can play a key forward looking role in integrated planning and long term sustainable and economic growth of city regions and mid to large cities (HLC, 2024). This can be done by identifying and empowering an existing agency or department, such as urban development authorities, that can coordinate with other sector agencies to develop a long-term vision for the city region and lead integrated urban, transport and economic planning for long-term growth of the area. Such integrated planning will help identify where the largest growth payoff are and will open opportunities for systematically managing peri-urban growth, undertaking town planning schemes and promoting transit oriented development. A key role of such a designated body will be to identify, prioritize and plan for critical infrastructure, services and investments, identify financing options, support partnerships with firms and workers, and build growth coalitions that can boost the performance of critical sectors and the region (box 3.1).

Box 3.1. Mandate of designated authority for integrated urban, transport, and economic planning

- In collaboration with relevant sectoral departments and key stakeholders, lead development of a vision of urban growth for the metropolitan area/region including peri-urban areas.
- Lead preparation of data-driven and evidence-based city-regional plan (including peri-urban areas).
- Identify strategic investments and their location that will transform the urban agglomeration.
- Identify infrastructure needed to improve connectivity between the city and peripheries and within the municipal boundaries.
- Identify key partners and support growth coalitions.
- Identify financing options including through public–private partnerships.
- Lead learning and innovation through partnerships with other cities.

Such integrated planning has been initiated in several cities such as Chennai, Pune, Ahmedabad, Mumbai Metropolitan

Figure 3.2. Tiered approach to spatial planning



Source: Team Analysis, 2025.

Region, and Greater Noida (box 3.2) and can be scaled up to other cities. In Surat, the government of Gujarat has set up the Surat Regional Economic Development Authority (SREDA), which covers Surat and five adjoining districts. The authority's mandate is to introduce economic strategy and planning considerations as a key input to master planning by identifying drivers of economic and job growth, accounting for city endowments and global economic trends, and quantifying the resources needed for identified projects.

All constituent development authorities and municipal bodies report to the SREDA, while also bearing responsibility for regional and transportation planning, and for regional marketing.

A high-quality, regularly updated, granular, and accessible national database on cities can support planning for urban

growth hubs. The last census was conducted in 2011, and systematic data on cities across states is not easily available. Given the complex nature of cities, with their unique structures, spatial patterns, and needs, spatially granular data can help identify the most productivity-enhancing interventions in cities. Such a database could be built on MoHUA's existing portal such as the AAINA dashboard and the India Urban Observatory (IUO) and be expanded in collaboration with states. It could inform urban policies and planning and support integrated planning in cities.

Strengthening human resources, skills, and competencies in urban planning and other key sectoral areas such as urban economics, urban design, transport, finance, and climate change are needed to strengthen the growth potential of cities (HLC, 2023). Most cities focus mainly on basic services delivery. While metro and million-plus cities have

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Box 3.2. Integrated planning in Chennai Metropolitan Area (CMA)

The Chennai Metropolitan Development Authority (CMDA) is currently engaging in a series of integrated urban and regional planning studies for guiding and coordinating land use, transport and infrastructure planning, and strategic economic growth in the long-term for the CMA. It has been undertaking a comprehensive effort to map out key production and employment centers, and the housing and transportation needs of workers, families, and firms for preparing the Third Master Plan of the CMA for 2027-2046. The mapping has identified all “clusters” of production and employment using secondary survey datasets, administrative data, establishment-related data, and data from direct service providers such as the Electricity Board. To inform future land use planning decisions, CMDA is also conducting studies to better understand job-housing-workplace patterns and mismatch. Analytics to understand the physical and social infrastructure demand and gap considering shifting economic trends, resilience (flooding, urban heat island, etc.) are also ongoing. In Parallel, Chennai Unified Metropolitan Transport Authority (CUMTA) has carried out a 50,000-household travel surveys, which will be utilized to inform the preparation of a Comprehensive Mobility Plan and the Third Master Plan, to achieve truly integrated land use and transport planning, as well as infrastructure planning and service delivery across the metropolitan area.

Source: City consultations, 2024; HLC, 2024.

access to urban planners, this is not the case for smaller cities (Tier 2 and 3). A study conducted by the Town and Country Planning Office and National Institute of Urban Affairs (NIUA) shows that while 12,000 town planners are required in state town and country planning departments, a few years ago there were fewer than 4,000 such sanctioned posts, half of which were vacant (Niti Ayog, 2021). A World Bank analysis of staffing level for grade A, B, and C in 21 large cities in Tamil Nadu found vacancy rates of 25-74 per cent.²² A dedicated municipal cadre focusing on urban planning must be established (HLC, 2023). Cities should undertake a review of organizational structures, staffing and skills gaps, mandates, and technology integration, and should modernize administrative systems to address shortfalls. Technology solutions could also be developed for enabling public participation and interagency coordination (NITI Ayog, 2021). There has been positive action in this regard, for instance under MoHUA’s Special Assistance to States for Capital Investment (SASCI) scheme, around 1000 urban planners were recruited in FY2023-24.

URBAN ECONOMIC MANAGEMENT, ENABLING BUSINESS ENVIRONMENT, AND SKILLS DEVELOPMENT

For cities to meet their full potential as growth hubs, they must work well for firms and workers and establish or assign an existing department as the nodal department for city economic development. Contrary to successful examples from cities globally, such as New York, Barcelona, and Singapore (annexes 3 and 4), city governments in India typically do not have the mandate or resources to develop an

economic vision for the city; to provide for business development services, skills training, and sector-specific infrastructure for fostering the local economy; or to attract investments to the city. In case an existing institution cannot undertake the required functions, states should consider establishing a City Economic Development Council (CEDC) in their cities. As noted in box 3.3, the CEDC (or an existing institution) would not only develop an economic vision for the city but also play a leading role in understanding the constraints facing firms, especially young firms and those in sectors with potential for growth, needs of businesses and advocating for them with relevant public agencies. It would also champion the needs of micro, small and medium enterprises that play a key role in providing employment opportunities in cities, as well as support the informal sector. The CEDC would have strong representation from stakeholder groups such as local businesses and industry (HLC, 2023). Membership and participation of the state industry department, transport authorities, skills development agencies, and constituent urban and local government units, including the Zilla Panchayat would be critical (HLC, 2023). Operationalizing the CEDC will require it to be adequately funded with an annual budget and resourced with staff with a solid understanding of the local investment landscape and regulations, and with expertise in economics, business development, policy analysis, market research, data analysis, and communication. The governance structure of the CEDC needs to be worked out by the states. For example, various options could be considered, such as subsuming the CEDC and its functions within District Planning Committees, establishing it as a department of the municipal corporation or development authority,

Box 3.3. Mandate of the city economic development council

- Develop an economic vision for the city.
- Conduct data collection and regular assessments of urban growth and the economy.
- Engage with businesses, workers, and relevant state agencies (such as the industry, transport, and urban departments) to gather inputs and build consensus on an economic vision and potential growth sectors.
- Identify major bottlenecks as well as enablers for businesses—incumbents and potential entrants.
- Work with relevant government agencies at city and state levels to resolve bottlenecks and advocate for the enablers, such as streamlining compliance processes, zoning regulations, critical infrastructure, and regulatory reform.
- Work with technical and vocational education and training and other skills providers to align training programs with the needs of local industry.
- Develop strategies to attract and support young and small businesses through mentorship programs, incubators, and business development services.
- Develop strategic plans to vitalize or revitalize select locations in the city.
- Define key performance indicators and monitor the outcomes.

Source: Team Analysis, 2025.

or having it as a purely advisory body that advises the municipal commissioner and other key decision-makers.

Streamlining business processes can enhance firm productivity. A city's economic success ultimately depends on the productivity of its firms and workers. The policy and regulatory framework that firms face not only impacts their productivity (for example, by affecting access to imported components and technology), but also guides their prior decisions on what to produce, the scale of production, and which markets to cater to. While this framework is determined mainly by national and state governments in many countries—for example, while tariff rates in India are set by the central government per its trade policy, labor laws are set by both central and state governments, and decisions on development of industrial parks are often taken at state level—cities can influence this framework.

An enabling local business environment can enable economic density. Firms tend to thrive in cities in which they can access industrial, commercial, or office space—infrastructure and services—that is well equipped for their needs, and for some industries, where they can locate near input and output markets, can access finance, and get business development support. A smart regulatory environment for businesses and simplified processes in which permitting, licensing, and other regulations are fit for purpose and not too costly to abide by and enforce can enable more formal economic activity. Single-window clearance ensuring one stop for setting up businesses, facilitating renewals for

permissions, mandating time-bound services, and providing customs services, etc. could be set up at metropolitan level for megacities and mid- to large cities, and at municipal corporation level for smaller cities. Digital infrastructure can be modernized to scale up this process.

Cities must take a leading role in supporting job growth through skills development. Human capital is a key driver of firm productivity and wages/living standards for workers. High-quality education and health services can increase human capital and make the city more attractive for skilled migrants. Opportunities to increase skills and the support for finding jobs during transitions can also enhance local labor markets. By ensuring that workers and their families have access to affordable housing and institutions that build and preserve human capital (e.g., schools, colleges, and skilling centers, and health care centers and hospitals), cities play a direct role in building a pool of quality workers that firms can draw upon. States/cities can facilitate partnerships between education institutions and the private sector to identify future job-growth areas, skills gaps, and training/education needs in relevant sectors. Odisha, for instance, has taken proactive steps through the flagship initiative World Skills Centre to equip the workforce with the skills required for future-ready jobs, ensuring that its labor market aligns with evolving global demands.²³

Support the informal sector, which accounts for 75 percent of urban employment in South Asia.²⁴ Many of these people work in public spaces or private homes. To promote

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economic inclusion of informal livelihoods, cities need to consider urban informal workers and their livelihood activities when formulating housing, land, zoning, and local economic development plans and policies. Measures include providing basic infrastructure services of water, electricity, and sanitation in informal settlements as well as public spaces; improving public spaces and sidewalks; constructing facilities to support livelihood activities such as community marketplaces, kiosks, storage and treatment facilities; and improving access to existing government livelihood programs. The Rural-Urban Transition Policy and the Jaga Mission are among the several reforms that Bhubaneswar launched to address housing and infrastructure needs for marginalized communities. In addition, cities should allow cooperatives, producer groups, and other organizations of informal workers to bid for public procurement contracts for goods and services provided by informal workers.

ENSURE AN ADEQUATE SUPPLY OF AFFORDABLE SERVICED LAND AND HOUSING

Indian states and cities are pivotal for facilitating efficient land markets and well-functioning housing markets. Access to affordable land and housing by firms and residents has emerged as a major constraint in economic development in Indian cities. Some 7.7 million people are affected by conflict over 2.5 million hectares of land, jeopardizing investments worth \$200 billion. Land disputes clog all levels of courts in India.²⁵ This is particularly holding back the land-intensive manufacturing sector in India and rapidly growing sectors such as logistics and warehousing. Urban

India also faces a significant housing deficit. Although government initiatives like the Pradhan Mantri Awas Yojana (PMAY) have made substantial progress towards the goal of affordable housing for all, demand remains acute, with projections reaching 31.2 million units by 2030.²⁶ Sizable housing demand in Indian cities is, in effect, being met by informal supply in the forms of slums, which house about half of the urban population. Addressing this rising demand for affordable housing is essential for ensuring that cities can attract and retain a diverse workforce, which is vital for sustained economic growth.

State governments should accelerate efforts to develop a modern and efficient land administration and management system. High fragmentation in land ownership, outdated and often conflicting land records, and lengthy land conversion processes make land assembly costly and inefficient. Governments should prioritize the digitization and integration of land records with cadaster maps, which provide a comprehensive picture of land parcels across cities, towns, and villages, and should keep them updated, thus enhancing land and property valuation practices (Box 3.4).²⁷ This will also help streamline the identification and acquisition of land by firms seeking to establish or expand business operations. Effective public land management is equally critical. Public lands in India are held by every level of government: municipal, metropolitan, state, center, as well as quasi-independent trusts and publicly owned companies. These public lands, if managed strategically and efficiently, can be pivotal in addressing land and housing shortages.

Box 3.4 .Integrated land administration and spatial planning in Indonesia

The World Bank's Integrated Land Administration and Spatial Planning Project (ILASP), with an investment of US\$653 million, is designed to transform Indonesia's land management and spatial planning systems. The project includes five key components: (i) Climate-Informed Spatial Planning; (ii) Strengthening Land Tenure and Landscape Management; (iii) Land Information System and Valuation; (iv) Large-scale Maps for Climate Action; and (v) Project management and Capacity Building.

The anticipated outcomes of this project are extensive. The ILASP will support the formulation of 500 detailed spatial plans at city level and integrated spatial plans for national strategic areas, enhance tenure security for vulnerable communities, establish a comprehensive multi-purpose land and geospatial information system, improve local revenue generation through mass land valuation, and strengthen governmental service delivery. In addition, by producing large-scale base maps and geospatial data infrastructure, the ILASP will enhance climate resilience, disaster risk modeling, and land use monitoring capabilities.

Through these strategic measures, the ILASP will significantly improve Indonesia's land management practices and spatial planning capabilities, optimizing urban land use, thereby boosting investors' confidence by enabling more informed decision-making and ultimately contributing to Indonesia's economic growth.

Source: World Bank ILASP Project Document, 2023.

For example, cities can leverage underutilized public lands to crowd in production of affordable housing by private developers. In addition, making the process for land conversion—from agricultural and industrial to other uses—simpler and more transparent, guided by integrated economic and spatial planning, can improve market efficiency.

Action on several fronts is needed to increase the effective supply of serviced land and affordable housing in cities.

First, a move from numerous and restrictive land-use categories to fewer, broad categories would make it possible to shift land more easily between uses across economic activities. Second, low and uniform floor space indexes (FSIs) need to be reassessed, as do building bylaws (such as rules on setbacks and minimum plot sizes).²⁸ In particular, FSIs should be commensurate with planned and prospective infrastructure upgrades rather than held back by legacy infrastructure. Especially when used with transit-oriented development (TOD) principles, higher FSIs could help make cities more compact, allow services-related growth drivers to better exploit agglomeration economies, and lessen some of the pressure on peri-urban development. Third, to address supply-side constraints of affordable housing, cities should consider measures such as relaxing restrictive zoning and development control regulations, streamlining planning and construction permits, implementing inclusive zoning policies, and introducing mechanisms such as a density bonus (additional FSI) as well as transferable development rights. These actions can enable private developers to offer a range of housing solutions, including ownership and rental options, catering to evolving demands.

Other measures to increase the availability of affordable housing include coordinating land use with infrastructure development and integrating informal housing supply into the overall market. Cities can improve existing infrastructure to support housing infill and redevelopment. Upgrading informal settlements by providing essential services such as

water, sanitation, drainage, solid waste collection, and public transportation is crucial. Efforts to address slum growth and improve unsafe, unsanitary conditions through enhanced housing should ensure that workers maintain access to labor markets. It is equally important for cities to coordinate land use with infrastructure development and to promote mixed-use, mixed-income communities strategically located near transit systems, facilitating access to jobs and amenities. These measures, along with continued support under the PMAY, will help meet the needs of a broad spectrum of residents, from new migrants seeking jobs in cities to executives and entrepreneurs spearheading urban growth.

More cities in India should leverage land-pooling tools to ensure land availability in peri-urban areas.

Land pooling is an innovative approach to securing land for urban expansion and constructing key infrastructure where landowners voluntarily contribute their land parcels to a common pool, which is then developed with modern infrastructure and amenities. Once the development is complete, the landowners receive a portion of the reconstituted land, which is often more valuable than their original holding. Voluntary in nature, land pooling in India offers a viable alternative to traditional land acquisition methods, providing several benefits such as increased transparency, cost-effectiveness, and inclusivity. Land pooling has been successfully implemented in cities like Magarpatta. However, to scale up its use, steps should be taken to introduce enabling state-level legislation with the flexibility to adapt to local context and experience, build trust with landowners, establish efficient conflict-resolution mechanisms, and develop the capacity of skilled planners and officials. Additionally, strategies to minimize delays in execution and improve financial viability are crucial, particularly in smaller towns where land value appreciation may take longer. While land pooling holds promise, other models such as leasing, and direct purchase (table 3.1) should also be enhanced to meet diverse urban development needs (annex 5).

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Table 3.1. Securing land access for infrastructure and public service delivery

Methods	Key features	Good practices
1. Direct purchase	Land is bought through private negotiation.	Ensure accurate and updated land records to identify rightful owners and avoid disputes.
2. Lease	Land is rented for a project, with periodic rent adjustments based on market value.	Include clear terms for rent revisions and project duration.
3. Public–private partnership (PPP)	Private entities build or develop government-owned land and operate for a set time before transferring it back.	Clearly define roles, responsibilities, and profit-sharing to ensure long-term success.
4. Tender	Landowners submit proposals for selling land at nominated prices after tender announcements.	Use transparent processes and proactive land planning to mitigate project risks.
5. Exchange	Land is exchanged for other assets of equal or greater value.	Ensure fairness in valuation and consider exchanges between public bodies.
6. Land pooling and readjustment	Property owners pool land voluntarily, share profits, or receive land back after development.	Promote voluntary participation and equitable profit distribution to build trust.
7. Joint development	Public and private entities collaborate, contributing land, resources, and expertise for mutual benefit.	Align public and private goals for coordinated infrastructure and development.
8. Easement	Allows access over/under land for utilities like pipelines or power lines.	Fairly compensate landowners for granting access.
9. Planning obligations	Developers contribute cash or infrastructure improvements to offset project impacts.	Tie developer obligations to tangible community benefits, such as traffic or environmental mitigation.
10. Compulsory acquisition	Government mandates land sale for public projects.	Ensure fair compensation and minimize community displacement.

Source: Team Analysis, 2025.

IMPLEMENT SOLUTIONS FOR EFFICIENT CONNECTIVITY

Good connectivity is essential for urban growth hubs in Indian cities as it enhances economic efficiency and competitiveness by reducing travel time and costs, enabling workers to access job opportunities and businesses to efficiently reach markets and resources. It fosters economic clustering, where industries and services benefit from proximity, leading to increased productivity, innovation, and knowledge-sharing. The Delhi–Mumbai Industrial Corridor exemplifies this by aiming to create smart and sustainable industrial cities with high-speed, high-capacity connectivity. Further, efficient transport systems enhance the quality of life by improving accessibility to education, health care and leisure, making cities more attractive for investment and talent. India’s growing economy has witnessed a rise in demand for transport infrastructure and services—efficient and reliable urban transportation systems are crucial to meet this demand.

While growing hubs typically face congestion around the world, efficient planning is vital for balancing demand for mobility and logistics with space constraints. Most high-growth cities experience heavy congestion,²⁹ but they typically put in place plans to manage such congestion by integrating economic, urban, and transport planning (Box 3.5). The concentration of flows allows the use of more space-efficient options like rail or bus corridors. The co-location of affordable housing and dense job centers with public transit facilitates a high public transit mode share for most citizens. The creation of highly walkable neighborhoods reduces the need to depend on motorized transport for many trips. And the development of consolidation centers enables more effective logistics.

Effective mobility and logistics in Indian growth hubs require a fine tuning of the approach to comprehensive mobility planning. While over 25 megacities and mid- to large cities in India have developed—or are in the process of developing—comprehensive mobility plans (CMPs), as for instance Kolkata Metropolitan Area, they are in many cases

Box 3.5. Green urban transport development in Tianjin, People's Republic of China (PRC)

The Tianjin Urban Transport Improvement Project (TUTIP) was a World Bank-funded initiative aimed at making transportation in Tianjin, PRC, greener and safer. The project focused on leveraging the Tianjin metro system and promoting walking and biking in the urban core. The project included components such as financing the redevelopment of streetscapes in identified districts, creating integrated pedestrian and bike networks with infrastructure investments like street pavement updates, drainage improvements, street facilities, and landscape improvements. Metro Access Improvement component financed selected intersection improvements and interconnection facilities (bike parking, bus connection/terminal, taxi connection, landscaping, and park and ride) at about 96 metro stations along existing Tianjin Metro Lines. Bus Terminal Development component financed the development of selected bus terminals, including pavement of terminals and bus stops, bus parking, car parking, bike parking, and service buildings, as well as equipment for bus operation within the terminals. The project also included technical assistance on sustainable green urban transport development, parking management improvement schemes, and analysis of economic activity in the non-motorized transport zones supported by the project.

Source: World Bank, 2015. TUTIP, Project Appraisal Document.

not backed by legislation, are more indicative than compulsory in implementation, and generally only partially integrate logistics aspects. Good practice, as showcased in cities like Delhi or Chennai, draws on detailed data and aligns the preparation of those plans with the city or metropolitan urban master plan to foster coordination and synergies and to focus on outcome indicators of accessibility and sustainable mobility. Further, international growth hubs integrate complete logistics plans in city plans, considering both inter- and intracity flows of goods, thus supporting infrastructure and services and accompanying policies.

Connectivity for growth hubs can focus on hubs and clusters. First, at city level, the concentration and flow of people need to be matched by its transit system to provide space and cost-effective mobility options and accessibility. This entails a hierarchical and well-spread network of integrated modes to cater to all types of demand—passenger and freight—with high-capacity systems on trunk routes for passengers and freight, availability of buses and para-transit modes within minimum walking distance, walkable neighborhoods, consolidation and distribution centers, and loading/unloading zones. Second, at cluster level, connections between cities require a dedicated focus to concentrate and support flows of goods, services, and people. Those can leverage on India's rapidly expanding highway network to drive growth, by integrating services like rapid, long-distance electric buses with quality bus terminals well located in such cities or by developing logistics hubs serving such clusters. The national and state-level rail systems can further strengthen cluster effectiveness by connecting city centers with rapid connections, for which national or state planning is required.

Operationalizing strategies require institutional mechanisms to articulate and implement policy and planning measures and allocate corresponding financing. While MoHUA supported such concepts for a decade, only a handful of cities in India have operationalized a unified institutional mechanism for transport. Megacities and metro cities need a high-capacity unified transport agency—for example, the Chennai Unified Metropolitan Transport Authority (CUMTA)—to ensure alignment across master plans and mobility plans; oversee resource allocations and system performance; and promote collaboration between various government agencies, private sector stakeholders, and community organizations to achieve common transportation goals, thereby fostering a more integrated and effective urban transportation system (annex 6). A state-level institution can support several small and medium cities of the state, such as the Directorate of Urban and Land Transport (DULT) in Karnataka. At the same time, urban local bodies need to be strengthened to plan, implement, and manage the diverse transport infrastructure.

The development of public transport at scale and pace in growth hubs requires dependable resource allocation and efficient use of resources. Public transport plays a major positive economic, social, and environmental role in growth hubs, but seldom recovers its financial cost in India like in other countries. Funding for public transport has traditionally been too limited to meet demand expectations, while affordability considerations have limited tariff increases, creating downward pressure on service delivery. All international growth hubs have recognized the need to allocate regular funding to public transport based on targeted levels of service delivery and efficiency. In India, Chennai has

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now adopted a Public Transport Service Contract with its operator public bus operator, MTC, with key performance Indicators and viability gap funding. An increasing range of cities have adopted gross cost contracting of bus services to bring in private efficiency in electric bus service delivery, thereby reducing service delivery costs. Further, reforms have been launched to integrate buses and para-transits either at transport department, state, or city level where special-purpose vehicles provide city bus services and e-rickshaws for first- and last-mile connectivity (Capital Region Urban Transport, CRUT, in Bhubaneswar). Additionally, investment in complete streets with walkable sidewalks and cycle lanes enhances safety and accessibility for all users and supports the broader goals of increased public ridership. Such solutions will be needed at scale to support cities as growth hubs.

FINANCING CRITICAL URBAN INFRASTRUCTURE AND SERVICES

Investing in urban infrastructure is crucial for spurring economic development in cities. A World Bank study estimated that \$840 billion, or 1.18 percent of India's GDP, is needed from now through 2036 for improving urban infrastructure and municipal services.³⁰ To ensure that cities have the capacity to mobilize resources to create, operate, and sustain urban infrastructure assets and services, a coherent effort targeting policy is required, with institutional and administrative reforms at all tiers of government (annex 7).

Central and state governments must continue to create an enabling environment for improving cities' creditworthiness. To foster fiscal autonomy among cities and inculcate marketability in urban infrastructure investments, there is a need to shift the focus from excess dependence on fiscal transfers to alternative sources of financing such as land-based fiscal tools, debt financing, and enhanced own-source revenue (OSR). Administrative and policy improvements towards OSR, such as improved records, mapping, collection efficiency, and progressive taxation policies linked to service levels, help boost investor confidence and foster private sector participation. To facilitate access to commercial financing, states need to establish a conducive regulatory environment for municipal borrowing and PPPs and put in place institutional mechanisms at central and state levels to enable large cities to access capital markets and contract the private sector. For smaller cities, states should aim to strengthen state-level financial intermediaries (FIs) to provide technical support on upstream project preparation

and help cities access municipal loans or pooled financing. A standardized debt management/capacity assessment framework to support cities in assessing their debt servicing capabilities and related risks is also recommended.

A multiyear "city infrastructure and investment plan," linked to the long-term vision of the city, should be developed.

For city regions and large cities, it should be prepared at the metropolitan and city levels with relevant capacity in place. To improve implementation capacity at city level, states can transition to a rules-based fund-allocation process (instead of a direct control system) along with performance-based indicators linked to results (especially economic factors). States can standardize fiduciary systems, as well as accounting norms and practices, such as reporting and disclosure of audited financial statements, comparable information, and a standardized integrated financial management system. To generate interest from the private sector, cities can engage in dialogue, with the state's support, to expose FIs/investors to the market potential and favorable government policies for financing urban infrastructure.

Cities have various financing options available for financing urban infrastructure (figure 3.3). The applicability of these financing options varies depending on city size and growth typologies (see chapter 2).

URBAN AMENITIES, LIVABILITY, AND RESILIENCE

There is a growing relationship between cities with strong climate resilience and mitigation plans, and their potential to attract and retain investment, business, and talent.³¹

Indian cities face an existential threat from climate change impacts, particularly from urban flooding and heating. Urban flooding-related losses are estimated at 0.5-2.5 percent of GDP. Megacities—many of which are riverine or in coastal areas—face staggering costs due to recurring flood hazard events and concentration of people and assets. An estimated 160 million-200 million urban dwellers in India are likely to be exposed to lethal heatwaves by 2030 (McKinsey Global Institute, 2020). Additionally, urban centers now account for 50 percent of national emissions, largely driven by energy and industrial sectors. The long-term viability of cities depends crucially on ensuring water sufficiency, reducing air pollution, and protecting against shocks and stresses from disasters and climate change. Putting in place city climate change action plans is an important first step. Cities should invest in early warning systems and emergency responses,

Figure 3.3. Financing options for urban infrastructure

City own-source revenue		Fiscal transfers	Commercial and private financing	Others
Tax revenues: property taxes, professional tax, entertainment tax Non-tax revenues: user charges, rental income, advertisement fees, investment income	Leveraging land assets (e.g., land value capture tools): FSI premium, development charges, betterment charges, tax increment financing, business improvement district, sale of land	Central: Central Finance Commission grants (e.g., new cities incubation grant), centrally sponsored schemes (e.g., smart cities mission) State: State Finance Commission, assigned revenue, state grants, etc.	Municipal bonds Scheduled commercial banks Blended finance PPPs / private capital Guaranteed loans (e.g., HUDCO) Loans from MDBs	State intermediaries (e.g., TNUDF) Municipal development funds Grants from MDBs (through central and state govt.) Corporate social responsibility grants
All types of cities	City regions/ Mid- to large cities	All types of cities	City regions/ Large cities with strong financial capacity with strong financials	Medium and small cities

Note: The above are some of the key sources of revenue for cities. The above typologies may not be fully exhaustive and may not fully account for some types of overlap between categories in some contexts. Tax increment financing involves borrowing; many forms of land value capture involve a PPP of some kind; PPPs often involve a form of indirect borrowing. HUDCO = Housing and Urban Development Corporation; TNUDF = Tamil Nadu Urban Development Fund; MDBs = Multilateral Development Banks.

which are “quick wins” and cost-effective investments saving lives and property. The Heat Action Plan prepared by Ahmedabad Municipal Corporation has helped avert more than 1,000 deaths annually.³² Cities should also integrate green infrastructure or nature-based solutions in their risk mitigation measures (box 3.6).

Access to basic education and to health services and amenities is important to attract and retain residents/workforce. Experience from developed countries highlights that vibrant, low-crime cities with public facilities and key amenities such as health and education services attract talent and boost economic growth. Indeed, cities deemed charming and beautiful were found to grow more quickly (Carlinio and Saiz, 2019). Investments in resilience, green design, walkability, and improved air quality can also help attract highly skilled workers to a city. Urban design boosts a city’s identity and creates a more attractive and sustainable business environment. From the ability to reflect a city’s history, culture, and character, to providing an aesthetically pleasing environment that promotes walking and cycling, urban design plays a key role in attracting not only tourists but also retaining a highly skilled workforce. Pedestrianization initiatives in the dense core city areas of Chandini Chowk in Delhi and the Bhadra Precinct in Ahmedabad have seen higher footfalls

and economic activity, better visual integrity, improved safety, and gains in local air quality. The urban space needs to be safe for all.³³

TOD is an urban design paradigm that has the potential to promote compact, dense, environmentally sustainable, and mixed-used development. Implementation of TOD around metro stations and transit corridors will benefit from mixed-income and high-density developments that have adequate green/open spaces and last-mile connectivity, as India continues to invest in mass transit systems such as metro rail, which is a sizable central scheme. Several cities such as Delhi, Ahmedabad, Mumbai, and Naya Raipur have TOD policies in place and others such as Bengaluru and Kochi are adopting such policies—and such adoption should be scaled up. A practical manual for implementing those policies has been prepared by Delhi.³⁴ Still, implementing TOD policies faces numerous challenges such as land acquisition issues, outdated zoning regulations, and governance and coordination issues (as discussed in this chapter and summarized in table 6.1 that also need to be addressed for successful implementation of TOD policies.

The quality of publicly provided network infrastructure is a critical input into production. Working with relevant state

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Box 3.6. Piloting nature-based solutions for urban cooling in Guangzhou, People's Republic of China

The city of Guangzhou, one of the largest and fast-growing cities in the PRC with a population of 19 million, demonstrates how cities can use nature-based solutions to mitigate urban heat and improve the living environment through developing three pilot sites in partnership with the World Bank. These solutions include:

Making use of natural cooling sources. Forests, lakes, and rivers serve as a natural cooling source. The city has identified and restored these cooling sources while preserving six major ventilation corridors along mountains, water systems, and open spaces. In addition, the city is using vertical greening on buildings as a simple, low-cost, and effective insulation, which reduces facade temperature and filters out glare and dust.

Restoring ancient local architectural climate wisdom. In the historic neighborhood, traditional building elements such as hollow walls, double-layered tiled roofs that insulate against heat, and flexible doors and windows that promote ventilation have been restored and reused.

Creating “breathing” urban fabric. The pilots have adopted urban design measures including widening some streets, decreasing the size of buildings, using light color coatings and rooftop greening, planting local trees, and increasing water features to cool the neighborhoods.

Source: Piloting Nature-based Solutions for Urban Cooling, World Bank, 2022.

agencies and private providers, cities should ensure that the location and design guidelines for strategic critical infrastructure such as power, transport, water supply and sewerage, solid waste management, and telecommunications are based on principles of resilience, sustainability, and the circular economy. City regional and master plans need

to be informed by flood-risk assessments and to integrate resilience and sustainability into their spatial-planning and development-control practices. Cities should also update design guidelines for buildings to incentivize green building and construction and to integrate resilience into critical infrastructure and services (table 3.2).

Table 3.2. Building regulations addressing climate risks adopted by some Indian states and cities

Type of risks	Example building regulations	States/cities where regulations are adopted
Floods: Restrictions on construction of basements or cellars	Basements or cellars shall not be permitted in low-lying areas without adequate drainage facilities to ensure drainage from the basement, and basements shall not be allowed in flood-prone areas. Construction of basements or cellars may be allowed by the Authority in accordance with the provisions contained in the development plan applicable to the concerned area.	Odisha Development Authority (Planning and Building Standards) Rules, 2020
Floods: Restrictions on buildings constructed in flood prone area	No land development or redevelopment shall be made or no building shall be constructed in a plot liable to flood or on a slope forming an angle of more than 45 degrees with the horizontal or on soil unsuitable for percolation or on area shown as floodable area in any town planning scheme or in sandy beds, unless it is proved by the owner to the satisfaction of the Secretary that construction of such a building will not be dangerous or injurious to health and the site will not be subjected to flooding or erosion.	Andhra Pradesh Building Rules, 2017
Floods and heat: Paved surfaces	Maximum of 50% of the total open space including marginal open spaces and common plot of a building-unit shall be paved. The remaining shall be permeable for rainwater percolation. At least 30% of the open spaces (setback area) shall be pervious. Use of grass pavers, paver blocks with at least 50% opening, landscaping etc. would be considered as pervious surfaces. The unpaved area shall be more than or equal to 20% of the recreational open spaces.	Gujarat Comprehensive General Development Control Regulations, 2017 Karnataka Model Urban Building Bylaws Andhra Pradesh Building Rules, 2017 Punjab Urban Planning and Development Building Rules, 2021

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Table 3.2. Building regulations addressing climate risks adopted by some Indian states and cities (*cont.*)

Type of risks	Example building regulations	States/cities where regulations are adopted
Green cover	For a building unit with an area of 10,000 sq m or more, an additional 6% area of the building unit shall be provided for thick plantation. This area shall be exclusive of the required minimum margins and common plot. The types of trees shall be selected such that they are shade-giving trees. A minimum of one tree for every 80 sq m of land should be planted and maintained. The existing trees will be counted for this purpose. Preference should be given to planting native species. Where trees need to be cut, compensatory plantation in the ratio of 1:3 shall be done and maintained.	Gujarat: Comprehensive General Development Control Regulations, 2017 Karnataka Model Urban Building Bylaws

Source: Team Analysis, 2025.

IV. IMPLEMENTING THE CITIES AS GROWTH HUBS FRAMEWORK

Previous chapters described key patterns of Indian urbanization, developed typologies of cities, discussed why Indian urbanization is under-leveraged, and proposed a framework and key building blocks that enable cities to serve as growth hubs.

This chapter presents a template (table 4.1) comprising specific steps that cities could take as they seek to realize their potential as hubs of economic growth and job creation. That is followed by illustrative case studies of selected Indian cities on their economic characteristics and potential. The exercise is useful for illustrating how the framework proposed in this report can be implemented, how solutions to the problems that hold cities back can be structured, and how a set of action points for the short, medium, and long term can be developed. More detailed guidance is in annex 8.

The strategies and options available to cities will vary depending on the typologies. For instance, city regions have a much larger and skilled labor pool and a more complex

array of institutional actors that shape urban economic growth than mid-to large or small cities. City regions and large cities also have greater capacity to access capital markets and contract the private sector compared to smaller cities which may need to rely financial intermediaries to access municipal or pooled financing. Hence cities and urban areas within each typology will have to identify interventions most conducive to their context.

KEY STEPS FOR CITIES ASPIRING TO BE GROWTH HUBS

Cities aspiring to be growth hubs would work in partnership with States to designate entity for integrated urban, transport and economic planning (particularly for city regions and mid- to large cities). Cities would also need to identify focal entity for undertaking city economic planning and locate themselves in one of the typologies proposed in this document. The following steps can then be undertaken by the designated entities (table 4.1).

Table 4.1. What should your city do if it aspires to be a growth hub?

	City regions	Mid- to large cities	Small cities	Transitional areas	Industrial urbanization
Responsible entity		Tasks to be carried out by the regional development authority or designated body		Tasks to be carried out by the city economic development council or designated body	
1.	Develop a baseline of the city's key features				
1.1	Assessing the city's economy				
	a) How much of the city and its surroundings constitute a single economy, i.e., an urban agglomeration? b) What are the main sectors, activities, and types of enterprises that drive the city's economy? c) How much direct and indirect employment do these drivers account for? d) What is the demographic and educational/skills profile of workers? e) Where are the city's main economic centers? f) Where do workers reside and how do they commute to work? g) How many industrial nodes are present near the city? Are they characterised by planned habitation in the immediate vicinity? Do these industrial nodes have a neighbourhood improvement plan? h) What are the top 3-4 marquee projects the city has undertaken to attract investments?				

IV. IMPLEMENTING THE CITIES AS GROWTH HUBS FRAMEWORK

Table 4.1. What should your city do if it aspires to be a growth hub? (cont.)

	City regions	Mid- to large cities	Small cities	Transitional areas	Industrial urbanization
1.2	Assessing connective infrastructure and mobility				
	a) How well is the city connected to other cities, nearby production hubs, and gateways? b) What are key features and challenges in using public transport in the city? c) Do airports, railways, and bus services have adequate capacity? d) Are there approved policies and plans for integrated mobility, city logistics, parking, and TOD?				
1.3	Assessing land and affordable housing				
	a) How much land is available with different government agencies in the city? b) What are existing mechanisms to access land for public purposes? c) Is there sufficient affordable housing for workers, including women workers? d) How does the city secure land for industrial, commercial, or housing needs?				
1.4	Assessing urban amenities, livability, and resilience				
	a) What are the top 3-4 marquee projects the city has undertaken to improve livability, amenities and urban services? b) What is the status of livability and access to basic services and amenities in the city? c) What is the status of well-being, nutrition, morbidity, and mortality of individuals in the city? d) What is the current accessibility to health care and sanitation facilities within the city? e) What are the current knowledge capabilities and status of access to schools, colleges, universities in your city? f) Does the city have quality public spaces and streets that follow good urban design principles? g) What are the key environmental and climate risks facing the city? h) How safe is the city, especially for women and children?				
1.5	Assessing infrastructure finance				
	a) Does the city have a city infrastructure and investment plan? Is there a pipeline of projects/investments identified by the city? b) What are the main infrastructure financing requirements of the city? What portion is expected to be met by public and commercial (external/private) financial? What is the gap? c) What are the different sources of financing and revenue of the city (fiscal transfers, OSR, debt, land revenue, private investments, etc.)? d) How much is the city able to spend on capex vis-à-vis the plan? e) What are the regulatory requirements/ gaps to commercial borrowing and PPP? What support is needed from the state on policies and guidelines? f) What is the debt management capacity of the city?				
1.6	Assessing governance structures and institutional planning capacities				
	a) What is the structure of urban governance in the city and its surroundings? b) Does the city have an institutional mechanism to coordinate economic, urban, and transit planning? c) Does the city have a tiered and updated master plan and land-use plans? d) Which are the key agencies involved in economic planning and development of strategies to promote economic activity? If these are at the state level, how do they interact with local authorities? e) What are the skill gaps and capacity constraints faced by the government in terms of economic, urban, and transport planning?				

IV. IMPLEMENTING THE CITIES AS GROWTH HUBS FRAMEWORK

Table 4.1. What should your city do if it aspires to be a growth hub? (cont.)

	City regions	Mid- to large cities	Small cities	Transitional areas	Industrial urbanization
2.	Demarcate Growth Boundaries and Key Actors				
	a) What is the boundary of the urban agglomeration to be considered for planning purposes? b) Who are the main institutional actors that need to be engaged at the agglomeration level?				
3.	Develop an urban growth vision for the agglomeration up-to 2047				
	a) What is the city's vision that incorporates competitiveness, livability, resilience, and inclusion? b) What are the main environmental and climate risks facing the city/region and how should they be considered as part of its growth? c) How can the city partner with other cities/infrastructure hubs to better leverage regional economies? d) What are the economic, urban, and transit investments needed to address the binding constraints facing the urban agglomeration but aligned with its vision? e) Is the existing urban, transport, and industrial planning aligned with the vision of the city/urban agglomeration?				
4.	Identify economic vision and strategy				
4.1	Understanding the business environment				
	a) How do city enterprises in key sectors compare with their regional, national, and global peers in terms of scale and productivity, and what are the markets they serve? b) What are the features of the city that enable economic activity in selected sectors? c) Do enterprises easily find workers with the necessary skills? d) What are the challenges faced by a city's enterprises in their operations? e) What are features that attract new investors to the city? Which do they find challenging? f) How can city enterprises improve their productivity? Are some local firms excellent performers, and if so, what enables them to excel in the same location as others? g) How are key industries addressing circular economy/clean production approaches? What are the key constraints? h) What is the value chain of the key economic drivers of the city? At what level of the value chain is the city functioning in that sector?				
4.2	Developing an economic vision and strategy				
	a) Does the city have an economic vision of what it would like to be famous for? b) What are the likely future economic drivers of the city (sunrise sectors)? What are likely sunset sectors? c) What is the type of support that enterprises expect from the government? d) Which are the agencies and levels of government with responsibility for delivering these "enablers"? e) How can cities ensure that their economic growth is inclusive? How can the informal sector be better engaged? f) How can industries better include clean production processes? g) How can cities work with enterprises to ensure that workers have the required skills?				
4.3	Identifying strategic manufacturing and services sector projects and investments				
	a) What are the industrial infrastructure and projects critical for supporting the city's economic growth? b) How should these be financed? c) Can they be structured to leverage private sector investments?				
5.	Identification of Strategic Projects and Investments				
	a) What are the priority investments for the city aligned with its vision? b) What are the financing options? c) What partnerships and capacity are needed to access the financing?				

Table 4.1. What should your city do if it aspires to be a growth hub? (cont.)

	City regions	Mid- to large cities	Small cities	Transitional areas	Industrial urbanization
6	Assess institutional levers, enabling policy interventions, and incentives				
	a) What institutional levers and policy interventions are required for cities to be growth hubs across building blocks—for example, to improve the functioning of land markets and better integrate economic, urban, and transport planning? b) How can existing governance frameworks be modified to provide a greater role for local authorities in economic matters? c) How can key actors, such as land developers, be incentivized to work with the intent of urban planners to improve the urban form of cities, discourage sprawl, and implement TOD policies?				
7.	Enhance capacity, establish partnerships, and conduct monitoring and evaluation				
	a) In which areas does the capacity of government officials engaged in economic, urban, and transport functions need to be augmented? b) What types of partnerships should be established for building this capacity? c) How can the city partner with educational institutions and other cities to enhance skills and innovation? d) What should be the key performance indicators that cities need to track as they aspire to become growth hubs? How should monitoring and evaluation systems be designed and implemented?				

Step 1. Develop a baseline of the city's key features.

The first step involves assessing the city in terms of its economy and key growth enablers, i.e., its transport infrastructure; land and housing issues; and urban amenities-, livability-, and resilience-related considerations. In addition, the governance structures of agencies tasked with key economic, urban, and transport functions and how they coordinate with one another must be clear. For example, if the agencies involved in economic functions are all at the state level, how well they coordinate with urban and transport authorities and other relevant departments such as industry needs to be understood. Similarly, the capacity to undertake various functions should be gauged. Finally, a city's finances need to be assessed.

Step 2. Demarcate urban growth boundaries and identify key actors.

As noted in Chapter 2, Indian cities of all sizes have expanded well beyond their administrative boundaries, encompassing multiple settlements covering urban and rural local government units that together can be thought of as comprising a single, large economy and labor pool. Moreover, the administrative and economic boundaries of cities do not coincide as 32 percent of the built-up area falls outside statutory cities. At the most basic level, the periphery serves as a source of

labor and agricultural products for the city, while the city provides employment opportunities, services, and infrastructure to the peripheral areas. A more dynamic economic relationship also exists. To better capture the economic boundaries, the economically relevant extent of the city periphery including its “urban growth boundary” must be defined (figure 2.4).

States and cities will also need to identify key public sector agencies and key actors such as firms, small and medium sized firm associations, worker associations and other stakeholders that would need to be engaged in integrated planning of the area. This would include bringing in private sector squarely into the frame, embedding firm considerations into urban management and developing growth coalitions. See, for example, figure 4.1 below for a mapping by Tamil Nadu of the various public sector agencies that influence economic development in the Chennai Metropolitan Area.

Step 3. Develop vision and growth strategy for the urban agglomeration

This step involves engaging with key stakeholders to develop a long-term growth vision for the urban agglomeration. Based on analysis done in step 1, it involves identifying priority economic, urban, and transport investments that are needed to address the binding constraints facing the urban area but aligned with its future vision. Here the lead agency would need to make decisions about key growth sectors,

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location of jobs and housing, how to better connect the center and peripheral areas of the city, how to avoid development in high-risk areas and grow in a way that enhances livability and resilience and reduces GHG emissions.

Step 4. Understand the enabling business environment and develop an economic vision and strategy.

Steps to understand how well a city's enterprises perform vis-à-vis national and global norms, what if anything holds them back from improving productivity (and encompassing possible constraints arising from difficulties in accessing appropriate located land, transport- and logistics-related bottlenecks, cumbersome business regulations, and absence of certain skill-sets, among others), and working with the private sector to tackle the identified constraints is crucial for developing an economic vision and developing a strategy for growth. Multi-stakeholder discussions spanning government, the private sector, workers, and experts are further required to decide on what might be sunrise and sunset industries and what types of enablers are needed—ranging from single-window systems to reduce the regulatory burden and investor promotion and facilitation support to encourage the entry of new investors to the city. Finally, special programs may be needed for certain groups of entrepreneurs and workers, such as common facilities for SME clusters; incubation hubs for encouraging startups; and interventions to address the factors preventing women from entering the labor force.

Step 5. Identification of strategic projects and investments.

Underlying any long-term growth vision and strategy will be a set of strategic projects—for example, a multi-modal logistics park, affordable housing for workers and their families, world-class skilling centers, investments in public transport and others. Developing a pipeline of such projects and working out a plan for their financing is a fifth crucial step that cities have to undertake. Governments will have to crowd-in private investment to fund some of these, and to ensure timely implementation.

Step 6. Assess institutional levers, enabling policy interventions, and incentives.

Ensuring that the building blocks of growth hubs are in place will require not just investments in projects in physical and

social infrastructure, but also the use of institutional levers and policy changes. Important examples are the need to improve the functioning of land markets by strengthening land administration, simplifying land conversion processes, and modernizing building bylaws; and to improve transport services through setting up effective, unified metropolitan transport authorities. It can also include soft support that can be provided to firms by government agencies (Tewari and Pillai, 2005). Similarly, institutional innovations that give a greater role to local stakeholders on economic matters need to be explored.

Step 7. Enhance capacity, establish partnerships, and conduct monitoring and evaluation.

Growth hubs can be demanding to manage and require high-level capacity in many areas. For example, implementing a town planning scheme or a TOD scheme successfully and without delay or designing an effective investor promotion program to invite outside investments to a city or a single-window system for local businesses all require capacities and skills that may be missing. Investments in building capacity, including through partnerships with other institutions, is crucial. Additionally, cities aspiring to become growth hubs should track key performance indicators and implement effective monitoring and evaluation systems.

The steps described above would require effort by city and state authorities, but that effort is sure to be well worth it, and it has demonstrable outcomes (box 4.1). Annex 8 describes in some detail what the steps on assessing a city's economy (step 1.1) and understanding the enabling business environment and developing an economic vision and strategy for growth (steps 2.1 and 2.2) would entail.

CITY EXAMPLES

A few cities (including Chennai, Bhuwaneshwar, and Indore) of which the first two were part of study field visits carried out for this report, were assessed on their economic aspirations, the nature of challenges that cities' enterprises/entrepreneurs face, as well as potential solutions to these challenges. The cities exhibit considerable diversity in their economic bases and aspirations. For instance, Chennai is an example of a mature, mixed city characterized by a strong manufacturing base. The city aims to leverage this as it targets more high-tech services and manufacturing activities. Bhuwaneshwar, in contrast, is an emerging contender for

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Figure 4.1. Public sector agencies that influence economic development in the Chennai Metropolitan Area

Planning	Visioning	Master planning			
	State Planning Commission	Chennai Metropolitan Development Authority			

Service delivery	Policies and promotion	Regulation/ease of doing business			Workforce development
	Guidance Tamil Nadu	Tamil Nadu Pollution Control Board	Directorate of Town Panchayat	Inspector General of Registration	Tamil Nadu Skill Development Corporation
	Tamil Nadu Tourism Development Corporation	Directorate of Industrial Safety and Health	Commissionerate of Municipal Administration	Tamil Nadu Fire and Rescue Services	Tamil Nadu Startup and Innovation Mission
					Tamil Nadu Labour Welfare Board

Infrastructure development	Industrial/ IT parks	Transportation	Utilities	Housing
	SIPCOT	Chennai Metro Rail Limited	Chennai Metropolitan Water Supply and Suraj Board	Tamil Nadu Housing Board
	TANSIDCO	Chennai Unified Metropolitan Transport Authority	Tamil Nadu Water Supply and Drainage Board	Tamil Nadu Urban Habitat Development Board
	TIDCO			
	ELCOT	Highways and Minor Ports Department	Tamil Nadu Generation and Distribution Corporation	Tamil Nadu Real Estate Regulatory Authority
		Transport Department	Greater Chennai Corporation	

Scope: Tamil Nadu Chennai / CMA

Source: Chennai Metropolitan Development Authority.

Box 4.1. Performance-based program to improve urban economic growth in Upper Egypt

The government of Egypt introduced a national program in 2016 to incentivize and facilitate integration of economic and urban management in the country. It committed to testing the model in a few governorates, and based on lessons learned from those experiences, evolve the model and introduce it to more governorates. The model comprised: (1) an incentive transfer from the national government on the basis of performance across indicators of economic and physical performance; (2) introduction of an economic council within governorate-level government; (3) establishment of one-stop shops for businesses, and simplification of select cumbersome regulations; (4) technical support to identify economic clusters of potential (e.g., solar panel production, agribusiness, textiles, and furniture), to assess their constraints, and to implement interventions to enable them to grow; (5) improvements in infrastructure and basic services; and (6) technical support to improve urban management and increase performance measurement. Over eight years, about 50,000 firms benefited from program interventions with an 85% satisfaction rate, and over 8 million people benefited from infrastructure investments with an 82% satisfaction rate.

Source: World Bank, 2024c.

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manufacturing industries—the city has a good skills base and requires an enabling ecosystem to move beyond its mineral hinterland and attract value-added activities. The third city, Indore, is well positioned to emerge as a manufacturing hub in the country with a strong manufacturing and logistics ecosystem and various green/sustainable initiatives under way.

These assessments are only illustrative and not exhaustive. More detailed analysis will need to be undertaken through detailed city-level diagnostics (such as that being undertaken for Bhopal and Indore—see Annex 8 for details).

Further, future analysis would ideally also cover issues pertaining to access to finance, education, and health. Chennai is currently preparing its Third Master Plan and as part of this, is developing an economic strategy for the city. Inputs from this strategy have been incorporated in our following assessment, including on the city's aspirations. For Indore, the assessment was developed through stakeholder consultations (with service and manufacturing firms, industry associations, and government agencies) and secondary research. Additional details are in annex 8.

The following tables summarize the illustrative assessments.

Table 4.2. Chennai

City aspiration	Build on its strong industrial base to transition into higher value-added manufacturing activities , and emerge as a strong provider of knowledge-intensive services. Specifically, the city can: • Move up the value chain in mobile phone and EV components by leveraging the links between industrial hubs and urban centers and promote corridor/cluster development. • Attract firms focusing on AI, R&D, FinTech. The city also seeks to foster inclusive growth by supporting the participation of females in the workforce and improving the quality of life through initiatives on transport and connectivity and affordable housing.
Enabling factors	• Good connectivity: Chennai is well connected through its international airport, railway station, and seaport (one of the largest ports in India) • The city is supported by a large manufacturing base: While Chennai is known for its automobile and auto-components industries, it also hosts diverse sectors like IT, petrochemicals, electronics, and textiles, attracting global players such as Ford, Hyundai, Microsoft, and Toshiba. • Emerging hub for warehousing and logistics parks:³⁵ Investments worth INR 10,000 crores from large players such as CapitaLand, Welspun One Ample Parks, and Mahindra Lifespace are being implemented. A Multimodal Logistics Park (MMLP) is also being developed by Reliance Industries near Chennai. • Chennai is rapidly emerging as a data center hub: Chennai's proximity to Asia-Pacific economies, coupled with lower latency times than several regional hubs, makes it a preferred location. Tamil Nadu's Data Center policy also offers fiscal and non-fiscal incentives for data center businesses. • Skilled workforce availability: Chennai has a high literacy rate (90.2%) and thriving education ecosystem housing six of India's top 100 engineering colleges and eight of India's top 100 universities.³⁶ • Conducive policy and regulatory ecosystem: Tamil Nadu is known for its ease of doing business and has various sector-specific policies. This will complement efforts by the Chennai Metropolitan Development Authority (CMDA) to better synchronize economic, urban, and transport planning as reflected in the preparation of the city's Third Master Plan. • Third Master Plan (2027-2046) for the Chennai Metropolitan Area (CMA) being developed: This will help resolve key urban issues as Chennai will be a major contributor to Tamil Nadu's vision of a \$1 trillion economy by 2030. • Skilled workforce availability: Chennai has a high literacy rate (90.2%) and thriving education ecosystem housing six of India's top 100 engineering colleges and eight of India's top 100 universities.³⁷ • Conducive policy and regulatory ecosystem: Tamil Nadu is known for its ease of doing business and has various sector-specific policies. This will complement efforts by the Chennai Metropolitan Development Authority (CMDA) to better synchronize economic, urban, and transport planning as reflected in the preparation of the city's Third Master Plan. • Enabling legislation for town planning schemes. The state has passed legislation that enables it to use land pooling/town planning schemes for assembling land for housing and infrastructure projects.

Table 4.2. Chennai (cont.)

Key challenges		Potential solutions
Low overall growth in the IT/ITeS sector relative to peer cities	• Concentration of IT growth in parts of Chennai Metropolitan Area (CMA) • Chennai has ceded IT investments to peer cities due to: ○ Less high-end social infrastructure (like international schools) for attracting global talent ○ Relatively lower capacity of the airport, including fewer connections to global destinations ○ Lower stock of developed land / real estate—fewer housing units being released annually; and less grade A office space to attract investors	• Transform underutilized zones into economic hotspots: Fastrack re-development of Ambattur and Guindy to house IT/ITeS firms and data centers • Accelerate the pace of IT development in North Chennai to promote balanced growth • Increase maximum permissible FSI in specific TOD zones and promote creation of grade A office space • Expand air connectivity to national and international destinations; accelerate the pace of development of Parandur airport to cater to this demand, while upgrading the existing airport
Transport- and mobility-related challenges	• Urban sprawl along transport corridors and the periphery has increased city traffic congestion • Over the past 15-20 years, there has been limited increase in bus services, leaving the city with just 23 buses per 100,000 population, compared with 100 in London and 300 in Singapore • 60-70% of suburban train stations are inaccessible by vehicles, limiting the effectiveness of the city's public transit system • Limited connectivity in peri-urban areas • Comprehensive TOD regulations required to spur growth	• Develop a parking policy and management plan, in alignment with the transit network and TOD with a focus on new growth areas such as new towns where IT/ITeS is expected to grow • Provide incentives for mixed-use developments along metro stations (promoting live-work-play concepts) • Accelerate implementation of city logistics policy covering peri-urban areas • Strengthen multimodal transport network, with a focus on last-mile connectivity
Land issues and limited housing Infrastructure	• Insufficient affordable worker housing near work centers • Limited success with TOD policy and land pooling under the Town Planning Scheme (TPS). . Additionally, in some cases, private developers have aggregated land plots and refused to pool land. • Low densification: Inadequate infrastructure to service additional population is a key concern for not increasing FSIs.	• Explore PPP model for industrial housing; scale up the approach taken in the Vidiyal Residency project by Tamil Nadu Shelter Fund • Scale up the Thozhi Working Women's Hostels to provide government-supported, safe accommodation for working women • Modify regulations to facilitate construction of affordable housing/ working women's hostels/ migrant housing, etc. • Provide incentives for rental housing through policy and regulatory interventions • Offer higher FSIs along metro corridors, major arterial roads and Mass Rapid Transport System corridors, accompanied by incentives for plot amalgamation so the FSIs can be absorbed • Develop capacity of officials to undertake land pooling and focus on effective communication with landowners to build trust

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Table 4.2. Chennai (cont.)

Sustainability	- Flooding and urban heat island effects are critical impediments - Resilience of power infrastructure is a challenge in the face of cyclones and flooding; additional focus is also required on green energy sources - Water: Depleted groundwater sources, seawater intrusion, and insufficient rainfall contribute to a demand-supply gap	- Climate resilient urban planning and implementation of the Climate Action Plan required - Adopt urban heat island mitigation measures through regulatory provisions - Create cyclone resilient transmission and distribution infrastructure which can ensure business continuity - Adopt the State Energy Conservation Building Code - Provide development control incentives for green energy adoption Adopt multifaceted approach to water resource management, including groundwater recharge, desalination expansion, rainwater harvesting, and reclaimed water usage, especially use of industrial wastewater.³⁸ Adopt urban heat island mitigation measures through regulatory provisions - Create cyclone resilient transmission and distribution infrastructure which can ensure business continuity - Adopt the State Energy Conservation Building Code - Provide development control incentives for green energy adoption - Adopt multifaceted approach to water resource management, including groundwater recharge, desalination expansion, rainwater harvesting, and reclaimed water usage, especially use of industrial wastewater.³⁹
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Table 4.3. Bhubaneswar

City aspiration	Serve as a gateway to Southeast Asia Develop Bhubaneswar-Cuttack-Puri (BCP) Economic Region⁴⁰ Build on strengths in resource- and employment-intensive sectors⁴¹ to emerge as a hub for marine food processing and for apparel Build on ecosystem for skills to emerge as a hub for IT/ITeS⁴²
Enabling factors	Emerging export gateway: Bhubaneswar is well-placed to serve as a major regional gateway to Southeast Asia with potential for port-led development Strong ecosystem for skills, R&D, and innovation - Odisha has a significant skills base that migrates to other states (like Tamil Nadu) in search of economic opportunities - It has a significant pool of industry-ready talent: 125,000 technical graduates annually from 773 training institutions - Presence of educational and R&D institutes of national importance like IIIT, AIIMS Bhubaneswar, Indian Institute of Management (IIM), Indian Institute of Technology (IIT), and NIT, World Skill Center Competitive cost of doing business - Availability of relatively low-cost labor - Stable and relatively low-cost power (enabled by power sector reforms and privatization) leading to lower cost of operations Emerged as an IT hub in the eastern part of the country: - Establishment of IT parks with plug-and-play facilities, customized incentives for IT companies, large talent pool, and low cost of living helped attract investments. - IT/ITeS majors like TCS, Infosys, IBM, Wipro, Mindtree, and Tech Mahindra set up their development centers in Bhubaneswar State government has implemented various urban reforms: Jaga Mission aims to make Bhubaneswar slum-free by upgrading informal settlements, improving governance, and securing land ownership for slum dwellers;⁴³ Odisha Rural-Urban Transition Policy aimed at providing essential urban infrastructure and services in the fast-growing rural areas adjoining cities,⁴⁴ etc. Enabling legislation for town planning schemes. The state has passed legislation that enables it to use land pooling/town planning schemes for assembling land for housing and infrastructure projects.

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Table 4.3. Bhubaneswar (cont.)

Key challenges		Potential solutions
Industrial infrastructure gaps	- Need for social infrastructure and common facility centers near industrial clusters Sector-specific gaps - Absence of large integrated textile parks - Absence of adequate cold storage capacity and packing houses for food processing sector - Despite being one of the major producers of inland and marine fisheries, there is only one major fishery harbor at Paradip - Lack of common facility centers for food processing	- Develop common effluent treatment plants to promote growth of textile clusters - Support logistics infrastructure to enable a smoother value chain in the textiles-apparel industry - PPP needs to be encouraged to: - Establish cold chain, common facility centers and pre- and post-harvest network infrastructure for food processing sector, including seafood - Social amenities in proximity to the industrial hubs to be developed - Develop alternate fishery harbors such as Puri, Ganjam, Gopalpur, Dhamra
Land and housing issues	- Limited land availability: Industrial development is spread across peri-urban areas such as Khordha–Cuttack–Jagatsinghpur (KCJ) - Land in peri-urban areas is needed for housing in BCP economic region - Limited availability of office space for IT/ITeS and other services sectors	- Explore utilizing land set aside for PCPIR (Petroleum, Chemicals and Petrochemical Investment Regions) for other industries - Expand current town planning schemes. - Utilize better land in city by considering repurposing of available public land and buildings, leveraging higher floor area ratio (FAR), examining scope for modernizing building byelaws, and streamlined approvals for constructing buildings for IT companies
Connectivity issues	- Low cargo-carrying capacity of Bhubaneswar airport (7,000 tons) while major manufacturing hubs such as Maharashtra, NCR, Tamil Nadu have capacity to handle more than 50,000 tons - BCP economic region suffers from inadequate connectivity, while access to the region's ports, notably Paradip, remains limited - Lack of adequate connectivity and mass transportation to IT parks and biotech parks.	- In addition to the development of the new passenger terminal for Bhubaneswar airport (planned), cargo capacity and connectivity (with national and international locations) should also be enhanced. - Fast-track implementation of Odisha Capital region ring road - Expand existing facilities to increase cargo-handling capacity at Paradip port and promote multimodal logistics parks to enhance overall supply chain efficiency - Include Paradip port connectivity under implementation project of a high-level partnership between Bhubaneswar, Cuttack, and Puri to improve inter-connectivity and market access - Develop high-speed corridors or flyovers to IT parks; launch dedicated bus services to and from IT parks
Water supply, sewerage, and other supporting infrastructure	- Storm water drain network of the city is almost 20-30 years old and unable to carry the peak load, leading to overflowing drains, especially during monsoons in residential, industrial and commercial areas and in knowledge centers - Bhubaneswar generates more than 90 megaliters per day of sewage without any organized collection and treatment method in place. Inadequate coverage of sewerage network service: 33% of the city	- Rejuvenate existing primary drains including covering and installation of filters - Modernize the stormwater drain network by incorporating ditches and detention ponds to help manage peak loads and reduce overflow during monsoons
Institutional challenges	- Bhubaneswar-Cuttack-Puri (BCP) economic region is represented by 3 development authorities whose actions are not coordinated - Labor regulation: New laws not notified - Lack of city data analytics (urban planning, challenges to local businesses, etc.) - Limited visibility among investors	- Explore setting up a regional development structure for BCP economic region and bring in Paradip port - Notify labor regulation reform to encourage labor-intensive industries - Establish data collection/ analytics cell to ensure data-driven urban planning and feedback collection from stakeholders - Supplement current investment-promotion efforts in target sectors

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Table 4.4. Indore

City aspiration	The Indore region can emerge as a major industrial hub for the country and also attract significant investments in IT/ ITeS and GCCs.	
Enabling factors	• Good connectivity: Its central location makes it a strategic distribution and logistics hub within the country. • Indore is well connected via national and state highways (like NH 3, NH 59, NH 59A, SH 27 etc.), rail, and airport to other parts of Madhya Pradesh and India • The city is improving air connectivity by building a new terminal to increase passenger capacity • Strong educational ecosystem: City hosts premier educational institutions, like the Indian Institute of Management (IIM) and the Indian Institute of Technology (IIT). Indore has about 60 engineering and more than 100 management colleges. • Thrust on sustainability: Indore has ushered in a clean and green revolution via various initiatives like: ○ Elimination of garbage dumps, 100% household-waste segregation, and conversion of waste to usable products, such as compost and fuel.⁴⁵ ○ The city has Asia's largest organic waste-based Bio-CNG plant (with a capacity of 550 tons per day)⁴⁶ ○ One of the world's largest 600 MW floating solar power park over Omkareshwar dam, with a 90 MW capacity, has been commissioned⁴⁷ ○ 60 MW solar plant at Jalud is being set up to power Indore's water supply⁴⁸ • Potential to be a start-up hub:⁴⁹ Low cost of rentals and operating costs (relative to bigger metros), incentives from the state's Start-Up Policy, presence of tie-ups with venture capital firms and sector-specialized mentors, and improving air connectivity. • Presence of diverse manufacturing base: The region hosts more than 29 industrial parks or zones across Indore, Ujjain, Dewas, and Dhar, along with six operational SEZs that cater to multiproduct industries, including major pharmaceutical and IT players. • Companies such as Aban Beverages, CP Paints, Bridgestone, Dabur India, Force Motors, Mahindra, IPCA laboratories, Lupin, Cipla, and many more contribute to the city's diverse manufacturing landscape, which includes textiles, food processing (savory snacks), electronics, automobiles, and pharmaceuticals • Indore has a National Automotive Test Tracks (NATRAX) • Enabling legislation for town planning scheme. The state has legislation that enables it to use land pooling/ town planning schemes for assembling land for housing and infrastructure projects.	
Key challenges	Potential solutions	
Limited availability of land for manufacturing	• Develop large-scale integrated industrial parks and townships with private sector involvement through enabling policies, such as a PPP policy for industrial parks and warehouses. • Explore large scale regional master planning and identify land for industrial needs • Utilize all mechanisms for land assembly, including land-pooling mechanisms and land acquisition for developing serviced land • Explore land-pooling mechanisms to develop serviced land	
Sector-specific industrial infrastructure gaps	• Plug-and-play facilities required for IT/ITeS and GCCs • Undersupply of common testing facilities for pharmaceutical units • Higher focus on agricultural warehouses; need Grade A & B warehousing facilities for key sectors (like pharma and auto); also need to ensure safety and security, and prevent damage to stored products • Need for dedicated air freight station to facilitate exports	• Consider policy interventions that would enable private real estate developers to create IT parks and high-quality office spaces with provision of plug-and-play facilities, including for start-ups • Establish common testing facilities for pharma and medical devices sector • Proposed Multimodal Logistics Park (MMLP) in Indore aims to support warehousing, transportation, and other value-added services⁵⁰ • Establish commodity-specific warehousing facilities, especially for pharmaceutical and automobile sectors • Develop additional cold storage facilities to support food processing and trade of perishables • Upgrade existing air cargo complex at Indore to ensure movement of high-value cargo

Table 4.4. Indore (cont.)

Transport and logistics infrastructure requires augmentation	- Poor condition of last-mile road segments within Dewas Naka, a prominent warehousing cluster, results in congestion delays and increased turnaround time - Need for mass transportation services between the industrial areas in Pithampur, Dewas, Mhow and residential areas in Indore	- Upgrade last-mile links at the prominent warehousing cluster of Dewas Naka - Fast-track development of Indore-Ujjain-Pithampur metro
Inadequate skilled workforce	- Limited National Assessment and Accreditation Council (NAAC)-accredited colleges in Indore - Companies face issues with hiring skilled workers for specialized domains such as design in the automobile sector	- Improve number of colleges with NAAC accreditation by encouraging institutes to apply for accreditation - Encourage investment in in-house training for specialized domains
Social infrastructure and housing gaps	- The social infrastructure facilities in Indore (health care, hospitality) are concentrated in certain locations (such as Vijay Nagar, AB Road) and not well distributed across the city - Limited industrial housing available	- Ensure better distribution of essential health care and hospitality facilities across the city by attracting new brands/chains - Increase availability of affordable housing through schemes such as PMAY - Undertake greenfield projects for industrial housing such as building houses for workers in industrial hubs⁵¹
Regulatory, institutional, and policy barriers	Disbursal of incentives takes time and availability of limited sector-specific incentives (for GCCs, electric vehicles, etc.)	Undertake review of industrial policy and introduce sector-specific incentives for electric vehicles, GCCs, etc.

EXAMPLES OF GROWTH STRATEGIES

Creative redevelopment

Creative Redevelopment refers to “activities that revitalize underutilized land, assets and outdated infrastructure through a systematic approach to unlock the development potential of a city through marketable interventions”. These can be achieved through all or a combination of (i) upgrading built systems for optimized use, urban form, and building resilience, (ii) improving mobility and accessibility, (iii) enabling regulatory and fiscal incentives, (iv) streamlining institutional frameworks, and (v) structuring innovative financing solutions (Creative Redevelopment Knowledge Framework note, 2024). The focus of Creative Redevelopment of Cities is to reimagine the city core through urban redevelopment and densification. Scarce, high-value, land and other assets within inner city limits of many Indian cities remain under-utilized—some lie vacant or dilapidated or are low density developments from decades ago. These need to be repurposed to meet economic, social and environmental objectives, which, in turn, could result in cities that are more efficient, green and livable. Cities should be dense with people and capital, not congested and crowded. This requires integrated economic and spatial planning covering transport and land-use. At present, there is insufficient focus on brownfield redevelopment and optimizing infill land in cities

(HLC Report 2, 2024). FSI in core brownfield areas are often lower than in peri-urban areas, which encourages unplanned greenfield development in the periphery. Preparing a land utilization policy, assessing and determining densities can identify untapped opportunities for infill and brownfield redevelopment, generate value and resources for cities (ibid.). This brownfield development of cities compliments the vision of developing Cities as Growth Hubs. For a more detailed discussion, please see separate Note on this topic.

Greenfield city development

Greenfield city development refers to building new cities from scratch on undeveloped land. A significant wave of new city building is occurring in the Global South, particularly in Africa, the Middle East, and Asia. Most new cities are envisioned as satellite cities to de-congest existing large cities, serve as new economic growth nodes, or promote technological innovation. However, greenfield city development also faces considerable challenges: it requires significant upfront investment to finance a broad set of infrastructure to attract businesses and residents away from well-established urban centres, and an efficient transport system to connect greenfield sites to existing urban areas. Research shows that new cities near major existing ones tend to be more successful due to access to established labour

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markets and infrastructure, benefiting from and contributing to greater agglomeration effects. International and Indian experience also highlights that the success of greenfield city development depends on strategic planning, robust governance, strong execution and management, and adaptability to evolving challenges. Governance structures for new cities may vary, but good practices underscore the importance of: (a) a degree of autonomy from existing political frameworks, enabling faster decision-making and innovative governance; (b) tailored administrative structure—some new city building started with a centralized governance structure for efficiency during early development stages, and decentralization to district or sectoral governance as the industries and population grow, and the economic base is stabilized and consolidated; and (c) starting with core function and a lean structure but incrementally expand to better respond to the evolving demands as city grows.

City clusters/system of cities

Many cities have become spatially connected with one another through urban expansion and the development of intercity transport infrastructure. The resulting agglomeration is said to comprise a system of cities or city cluster. City clusters can host large numbers of people and firms that are often highly diverse, generating large flows of goods, services, and ideas that make them uniquely vibrant urban areas in the region they are located in. City clusters offer

opportunities for firms and workers to benefit from enhanced agglomeration effects. This can happen through several channels that spread over a larger population the fixed costs of urban infrastructure, such as suburban rail networks and wastewater treatment plants; generate a thicker labor market for workers; and create opportunities for firms to exploit more efficient input–output linkages in the city cluster with a wider variety of industries, both upstream and downstream. City clusters also offer opportunities to invest in better environmental protection. If individual cities compete for investment, they may neglect the environment by either relaxing entry for polluting industries or failing to regulate the indiscriminate use of natural resources. A regionwide approach toward planning can help urban clusters develop in an environmentally sensitive and sustainable manner.

Realizing the full benefits of city clusters clearly requires good transport connectivity between their constituent cities and peri-urban areas. It also requires an appropriate governance framework. All too often, clusters suffer fragmented governance, with multiple administrations coexisting within a cluster. This can have negative consequences, as firms located in city clusters are more likely to report obstacles to business from regulatory and infrastructure factors, among them difficulty in access to land and licenses and connecting to electricity and telecommunication services, as well as transportation woes (ADB 2019).

V. DESIGN PRINCIPLES FOR A NATIONAL SCHEME TO PRIORITIZE URBAN GROWTH HUBS

A national program could play a catalytic role in enabling Indian cities to realize their economic potential. It could be structured with the objective of enabling growth in more and better jobs and/or a subnational GDP-equivalent. Given finite public resources, and the fact that all classes of Indian cities appear to be underleveraging the benefits of agglomeration, it is important to answer two key questions: Which cities could be the focus of a national program? And: How could the scheme be implemented? These questions are not easy to answer, but international experience provides pointers, starting with how to approach prioritizing cities.

WHICH CITIES? PRIORITIZING URBAN GROWTH HUBS

Economic growth requires vibrancy in all types of cities. It depends not just on one or two large cities performing well but also on well-functioning market towns that specialize in marketing and distributing agricultural produce as well as small and midsized cities, some of which may produce many types of goods and services while others specialize in a few products.⁵² This suggests that a national program could invest in all major types of cities. (Annex 9 discusses why this does not contradict agglomeration economies, which are sometimes taken to imply that because larger cities are more productive, it is best to focus investment on them.)

More generally, the allocation of public resources should prioritize cities with greater economic potential and where more people are moving. International and Indian experience indicates that cities with the following characteristics tend to attract more people and grow faster: better connectivity to major markets; better connectivity to major transport networks; larger pools of (skilled) workers; lower commuting and living costs; more employment in formal/larger manufacturing and/or services; the presence of large, anchor firms; and more young establishments.

Any framework for the spatial allocation of public resources must be kept flexible and informed by assessments of how the urban system is evolving. The following approach is proposed for prioritizing urban growth hubs.

100 CITIES TOWARD A \$30 TRILLION ECONOMY BY 2047

For India to meet its growth aspirations, the central government could, first, enable growth in the 100 highest economic potential or performance cities of varying types (annex 11). The prioritization criteria could be simple, for instance, one proxy for economic performance like working population. Alternatively, it could be a combination of a small number of indicators of economic potential and performance. One approach to measuring economic potential in India is the Economic Potential Index.⁵³ It combines five indicators for which there is robust evidence of association with local productivity: measures of market access, economic density, urbanization rates, human capital, and local transport connectivity.

Second, a regional balance criterion could be incorporated. Given the importance of enabling growth across regions of the country, adjustments to the top 100 could be made to ensure, for example, a minimum of one city per state is included in the top 100. Likewise, an adjustment could be made to ensure that at least one of each major type of city of importance to India's system of cities is included.

Third, criteria indicating a commitment to growth could be included. The willingness of governments to undertake necessary reforms linked to the India urban growth hubs framework, such as the regulatory and governance framework pertaining to economic, land, and transport planning at local levels could help increase the impact of public resources.⁵⁴ Criteria could include one or more of the following:

- Commitment to integrated economic, urban, and transit planning—*potential indicators*:
 - Designation or establishment of a metropolitan authority for integrated regional urban, transport and economic planning.
 - Development of an integrated metropolitan plan, and municipal and local area plans based on TOD principles among others.
 - The city has a long-term vision document and pipeline of investments identified aimed at triggering economic growth.

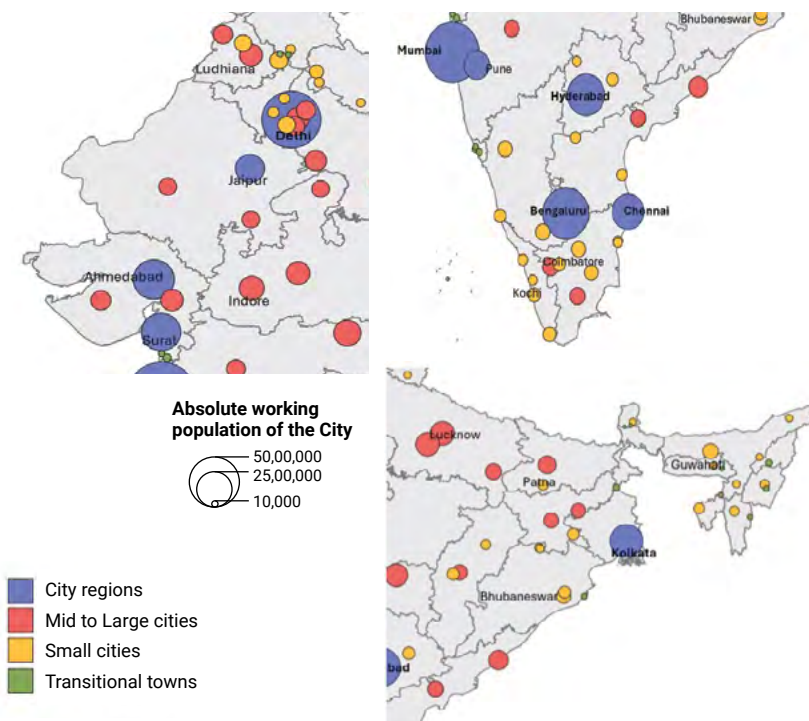
V. DESIGN PRINCIPLES FOR A NATIONAL SCHEME TO PRIORITIZE URBAN GROWTH HUBS

- Commitment to economic growth and coalitions—*potential indicators*:
 - Establishment of a city economic development department.
 - Establishment of an active public-private dialogue mechanism through the city economic development department.
- Commitment to enhance own-source revenue (OSR) and adopt innovative finance—*potential indicators*:
 - Increase in capital utilization ratio (capital expenditure / (opening balance + revenue surplus + capital receipts)).
 - Revenue surplus maintained over a period.
 - Increase in percentage share of OSR to total revenue.
 - Financial statements are audited and disclosed publicly.
- Commitment to address regulatory barriers such as land issues and onerous doing business regulations—*potential indicators*:
 - Proposal to address rural–urban transition and improve management of urban peripheries.
- Effort to address industrial urbanization in the state—*potential indicators*:
 - Setting up of a database to document unplanned industrial sites in the states.
 - An industrial estate plan that includes plan to systematically develop surrounding area.

States should be asked to provide a narrative about their vision and goals. While selecting one or a few indicators can be challenging, simplifying the number of indicators as much as possible can help increase the transparency and credibility of the selection process.

Map 5.1 presents one set of 100 cities emerging from such a prioritization, based on: (1) a simple economic criterion—working population; and (2) incorporation of a regional inclusion adjustment. The proposed approach to promoting growth hubs takes the top 100 cities by state/ union territory with the highest working population⁵⁵ (map 5.1). A minimum of one city from each state and union territory is taken, up to a maximum of six cities totaling 100 nationally. In addition to city regions, rising other urban area typologies are also reflected.

Map 5.1. Top 100 cities by state/union territories with the highest working population, categorized by typology



Source: Based on Census 2011, state boundaries from Survey of India, 2022 and sorted by typology using establishment data from Economic Census 2013-14.

V. DESIGN PRINCIPLES FOR A NATIONAL SCHEME TO PRIORITIZE URBAN GROWTH HUBS

Even if the ambition is to enable 100 growth hubs, support could be phased in over time. Starting with a first subset of cities—for instance, 25 of 100—offers two advantages. First is the opportunity it offers for learning from the start-up phase for a smaller set of cities and applying that knowledge to the start of support for a larger set of cities. Second, it may enable larger amounts of support to each city if phased

in over time. Both are important advantages, especially when considering lessons learned from recent urban missions in India (box 5.1). Most notably, insufficiently large funding seems to have been a constraint in the past, as were political economy considerations—one reason why including the third criteria above can help mitigate the risk of insufficient commitment to reforms.

Box 5.1. Lessons learned from central-level missions in the emergence of growth hubs

The central government has provided funding support over the last 20 years to cities in the form of a succession of “urban missions.” These programs provide assistance to state and local governments to meet their urban development and resilience challenges. They provide important lessons learned regarding the factors that determine the extent to which the center can successfully upgrade urban infrastructure and strengthen municipal management through fiscal transfers, bolstering cities’ development as growth hubs.

The Jawaharlal Nehru National Urban Renewal Mission (JNNURM, 2005–2012), for instance, focused on improving urban infrastructure and services, emphasizing water supply and sanitation. The Atal Mission for Rejuvenation and Urban Transformation (AMRUT, 2015–2023) aimed at improving water supply, sewerage, and green spaces in 500 cities. The second phase of the program (2021–2026) focuses on water efficiency. The Smart Cities Mission (SCM, 2015–2023) targeted economic growth and infrastructure improvement through smart solutions across 100 cities.

Political economy constraints have been the key limiting factor in these missions’ partial success. National missions have not been very effective in introducing changes to the institutional structure for urban management within states, that is, substantial state–ULB devolution, or in enforcing related conditionalities. Under JNNURM, for instance, ULBs faced challenges in implementing the 23 reform conditions, including state and local reforms, tied to funding. The conditions did not sufficiently incentivize institutional changes and none of the states met all targets.

Further, missions have been unable to overcome significant capacity gaps at municipal level. Institutional capacity at ULB level was limited, affecting the cost-effective implementation of AMRUT projects. The mission tried to address this through service-level improvement plans but faced challenges in maintaining coordination among institutions. SCM sought to navigate these issues through the creation of special-purpose vehicles for project management. However, the joint ownership structure between ULBs and states limited their autonomy, leading to fragmented responsibilities and delayed project implementation. Efficient implementation arrangements, which seek to minimize the fragmented division of municipal service responsibilities, are therefore important.

Funding modalities have similarly constrained execution. JNNURM saw delays in funding releases due to conditionalities, particularly towards the end of fiscal years, complicating project completion. AMRUT financing relied heavily on state and ULB contributions, with mission funding covering only a fraction of sectoral needs. Sustainable financing was a challenge, with inadequate tariffs and cost recovery. SCM funding required matching contributions from states and ULBs, with additional funding from PPPs and private capital. However, challenges in mobilizing these funds resulted in slower project implementation and incomplete financial leveraging targets. Going forward, these programs could therefore minimize counterpart funding requirements to level the playing field among states.

In addition, urban missions’ overall funding levels have been very small relative to total urban investment needs. Past programs were very ambitious but provided very limited financial incentives. It is therefore potentially more strategic for urban missions to maximize impact by financing fewer transformative large projects that shift urban development to an alternative trajectory through demonstration effects, as opposed to focusing on universal coverage.

Finally, issues involving land acquisition slowed project implementation in several instances. Land acquisition during JNNURM was a significant challenge, contributing to project delays. The lack of a clear procurement policy exacerbated matters. Land availability was a prerequisite for project approval under AMRUT, yet delays due to land acquisition were reported, particularly in states like Odisha. Delays in land acquisition similarly impacted SCM projects, especially in green-field developments.

V. DESIGN PRINCIPLES FOR A NATIONAL SCHEME TO PRIORITIZE URBAN GROWTH HUBS

For India to meet its growth aspirations, the greatest returns are likely to be in the largest cities. However, for the often-neglected transitional areas, namely, small towns and census towns with no specialization, the central government could set up a special program focusing on the top 100 transitional areas for technical and funding support (annex 10).

HOW COULD THE SCHEME BE IMPLEMENTED?

A national scheme to enable growth hubs could fund three components.

First, it could fund an annual performance-based transfer to subnational governments computed on progress against pre-agreed performance targets. Performance indicators could be unified for all participating cities and could comprise an appropriate output or outcome indicator reflecting the India urban growth hubs framework (see figure 2.4). Alternatively, a menu of indicator options could be developed with flexibility for each city to choose a subset of indicators, most relevant to its constraints and the types of interventions expected to deliver faster growth. The implementing entity could be uniform across scheme growth hubs or could vary by growth hub depending on *de jure* and

de facto differences in institutional capacities across growth hubs. An institutional analysis of state and local institutions can inform this decision.

Second, as changing governance and management practices can be difficult, the national government could fund substantial technical support for participating cities. This is a good practice widely adopted as part of urban performance grant programs worldwide. It also reflects a response to the lesson learned on weak capacity as a constraint to the implementation of previous national urban missions (box 5.1). Such capacity support would ideally comprise a demand-driven component and be oriented to the priority constraints and performance metrics included in the program.

Third, such a scheme would fund the development of a robust performance management system (box 5.2). At a minimum, this would involve developing the annual performance metrics, their measurement, and verification protocols. However, the system could be developed to enable more regular feedback loops so that it offers participating cities and other tiers of government a decision-making tool that could inform management decisions.

Box 5.2. International experiences implementing performance grant systems to enable growth hubs

Four relevant examples are Australia, the European Union (EU), the PRC, and Egypt. In Australia, performance-based grants financed by the federal level provided an incentive to subnational governments to implement pro-competitive economic reforms in 1995–2005. In the EU, all regional development investments must be aligned with policy objectives defined at EU and national levels, and the achievement of targets is incentivized through a performance reserve equivalent to 6 percent of the budget.

In PRC, subnational governments compete to achieve national planning priorities, which creates a strong incentive to increase business activity and the value of land in their area. Promotion of local government leaders is also linked to their performance on economic outcomes. In Egypt, the World Bank is supporting the government to pilot a grant system to promote competitiveness policies and investment in the lagging region of Upper Egypt by focusing on improving basic infrastructure and services, local government capacity for market-oriented private–public dialogue, improved government to business services, and industry- and sector-specific capital investments.

Source: Subnational Competitiveness Grants Guidebook. 2022. World Bank.

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

This chapter provides a summary of the key actionable recommendations that will be relevant for many cities and urban areas in the short term (ST), that is, 1-2 years, and the medium term (MT), that is, 3-5 years. An exercise will be needed in each state/ city context to define a few priority interventions.

Table 6.1. Implementation roadmap

Key issue	Recommendation	Responsible agency	Time frame
Central level actions			
Existing planning guidelines require revision.	Update Model Building byelaws and Urban and Regional Development Plans Formulation and Implementation Guidelines (URDPFI) to address integrated urban, transport and economic planning and sustainability.	MOHUA	ST
Need for higher level action to support states and cities on urban and regional planning.	Re-purpose Town and Country Planning Organization (TCPO) to create a hub which can provide expertise / handholding support to cities (similar to “Urban Renaissance Agency” in Japan). This hub should also have adequate representation from relevant Ministries (like Railways, Roads, Commerce and Industries, Ports etc.).	MOHUA	MT
Growth strategies for different types of cities need to be developed and action incentivized.	Set up a program including funding to support urban growth hubs and develop cross learning between city typologies on economic growth strategies.	MOHUA	MT
Centralized data repository for cities needed.	Set up high quality, regularly updated and accessible national database on cities.	MOHUA	MT
Unplanned urbanization around industrial areas; urban settlements inadequately integrated into industrial planning.	Develop program on “Industrial Area Urban Planning.” Hand holding support to be provided through reformed TCPO. Harness knowledge through partnerships with centers of excellence and education institutions.	MOHUA and MCI	ST-MT
State/city level actions			
Develop typology specific strategies and investment priorities			
Urban peripheral growth; mismatch between administrative and labor markets	Demarcate administrative areas most relevant to the city/relevant urban agglomeration, namely “urban growth boundaries.”	State/ Cities	ST
	Put in place rural-urban transition policy (Odisha good example)	States	MT
ULBs have narrow mandates and capacity; key decisions lie with line departments; thin technical capacity and tax base	Identify and empower key actors to lead the change specifically for (i) integrated urban, transport and economic planning, and (ii) economic visioning and planning; strengthen growth coalitions.	State/ Different City Typologies	ST-MT

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

Table 6.1. Implementation roadmap (*cont.*)

Key issue	Recommendation	Responsible agency	Time frame
Variation in capacity, resources growth drivers (e.g. access to markets, skill base, connectivity) between city typologies	Prioritize interventions and investments with a focus on relevant “urban growth boundaries.”	Different City Typologies	MT
Address most critical cross-cutting constraints			
Governance, planning, and institutional capacity			
Limited availability of trained urban planners at state/city levels	Fill priority vacant positions; scale up training and certification of urban planners	States in partnership with educational institutions	ST
Absence of integrated economic, urban, spatial, and transport planning at metropolitan and city levels	Designate existing or new department to lead integrated economic, urban, transport and logistics planning for metropolitan areas, including peri-urban areas; allocate staffing and budget. Systematize master planning: three tiers for city regions and mid-to large cities and two-tier planning for smaller cities.	States in collaboration with city development authorities and municipal corporations	ST
Limited availability of official data on cities resulting in challenges in measuring gaps, setting service standards, and putting in place performance management systems	Establish state-level multi-sectoral database on urban systems (including spatial, economic, demographic, land, resilience and other data)	States	MT
Strengthening cities’ institutions on economic development			
Absence of city-level focal department for economic visioning and planning	Designate existing agency/or establish City Economic Development Council (CEDC) or equivalent focus agency; clarify mandates, empower and allocate adequate staffing and budget	States/cities	ST
Cumbersome licensing, permitting and business processes	Set up city level one stop window for all clearances for setting up businesses; advocacy with central and state agencies regarding business regulations, tax/subsidy related issues	Cities	ST-MT
Lack of business development services	Partner with relevant agencies (SME development agencies, educational institutions, etc.) and private sector to address develop skill development programs, incubation hubs, common facility centers etc. for priority sectors	CEDC/equivalent agency	MT
Access to land and affordable housing			
Cumbersome/outdated land-use regulations and building bylaws	Relax restrictive zoning and development control regulations; streamline and simplify land-use conversion processes	States/cities	ST
Outdated land records and cadaster system	Establish a single land information system (ideally as part of comprehensive urban database); update land cadaster system; identify vacant land that could be repurposed; establish public land inventory.	States/cities	MT
Lengthy land assembly with difficult process of resettlements and compensation	Introduce land-pooling schemes for voluntary land development and ensure effective implementation through building capacity of planning officials, introduction of grievance redressal mechanisms. Enhance creative redevelopment through PPPs.	States/cities	ST-MT

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

Table 6.1. Implementation roadmap (cont.)

Key issue	Recommendation	Responsible agency	Time frame
Transport			
Lack of legislative backing for transport plans	Align comprehensive mobility and logistics plans with development plans/master plans to enable synchronization of land use and transport planning; invest in high-capacity system on trunk routes and flexible system such as buses for job accessibility.	States/metropolitan areas/cities	ST
Multiplicity of institutions with overlapping functions and weak capacity	Capacitate unified transport authorities in metropolitan areas (like CUMTA in Chennai) or state institutions (like DULT in Karnataka); clarify institutional mandates and strengthen coordination.	States/cities	MT
Limited funding resources and performance management for public transport affect service quality	Provide regular funding to public transport based on targeted levels of service delivery and efficiency (like the Public Transport Service Contract in Chennai); integrate buses and para-transit modes.	States/cities	ST-MT
Infrastructure finance			
Poor implementation and absorptive capacity of available funds	Improve cities' capacity to prepare pipeline investment projects; mainstream performance-based fiscal transfers linked to results; strengthen state-level financial intermediaries to provide technical support to small cities for project preparation, and help them access municipal loans and pooled bonds	Center/states/cities	ST-MT
Lack of staffing and weak financial management capacity undermine investor confidence	Establish a dedicated municipal cadre with technical and accounting staff; standardize fiduciary systems, accounting norms and practices	States	MT-LT
Lack of interest for private capital participation and mobilization	Regular dialogues with financial intermediaries/ investors to raise awareness of market potential, government policies, and available revenue streams to support infrastructure financing	States/cities	MT
Urban amenities, livability, and resilience			
Insufficient capacity for climate and disaster preparedness	Update infrastructure design guidelines and building codes to incorporate climate adaptation and mitigation measures; prioritize nature-based solutions or green infrastructure	States/cities	ST-MT
Lack of policies to promote compact development and contain urban sprawl	Put in transit-oriented development (TOD) policy and implementation guidelines to enhance densification, mixed land use, and walkability.	States	ST
Limited access to adequate infrastructure and services	Increase access to reliable water, waste management, sanitation, electricity and digital services; improve access to and quality of education and health services to both increase human capital and attract and retain talent	States/cities	ST-MT

Note: CEDC = City Economic Development Council; CUMTA = Chennai Unified Metropolitan Transport Authority; DULT = Directorate of Urban and Land Transport; ST = short term; LT = long term; MT = medium term; PPP = public-private partnership; TOD = transit-oriented development.

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

Table 6.2. City typologies: characteristics and available options

	City regions	Mid to large cities	Small cities	Transitional areas	Industrial urbanization
Geographical coverage	Mega cities and their contiguous urban agglomerations	Mid to large cities and their contiguous urban agglomerations	Small cities	Small towns and census towns	Varies depending on SEZ and estate size and can include multiple cities of different sizes linked with corridors
Population range	3 million and above	Population between 1-3 million	0.1-1 million	Population below 0.1 million	Varies
Examples cities	Mumbai, Delhi, Kolkata, Chennai, Hyderabad, Bengaluru, Jaipur, Ahmedabad, Surat, Lucknow, Indore	Coimbatore, Indore, Patna, Varanasi, Visakhapatnam, Kanpur, Ludhiana	Tirupur, Cuttack, Mangalore, Gaya Agartala	Doda; Farooqnagar (census towns); Amalner	Hosur (TN), Manesar (HR), Pimpri-Chinchwad and Talegaon (Pune), Bengaluru Electronic City (Karnataka), GIFT city (Guj) DMIC Corridor: a) Dholera Special Investment Region (GJ) b) Shendra Bidkin (MH); NOIDA SEZ (UP); SEEPZ SEZ (MH); Manikachan SEZ (WB)
Enabling institutions	Regional Development Authority Eg: MMRDA, NCRPB	Development Authority/ Municipal Corporation	Municipal Corporation/ Municipal Council	Municipal Corporation/ Municipal Council/ Nagar Panchayat/ Village Panchayat	SPV or designated urban or industrial development authority
Planning system	Put in place 3 tiered plans at the regional, city and micro-levels.	Put in place 3 tiered plans at the regional, city and micro-levels.	Put in place 2 tiered plans at the city and micro-plans	Put in place 2 tiered plans at the city and micro-plans	Master plan for SEZ/industrial estate must include integrated planning of surrounding urban area
Economic development and business environment	Single window clearance ensuring one stop for all clearances for setting up business Digital infrastructure	Single window clearance ensuring one stop for all clearances for setting up business Digital infrastructure	Single window clearance and access to unencumbered, serviced land with access to connective infrastructure (road, rail, airport, port) Digital infrastructure	Single window clearance and access to unencumbered, serviced land with access to connective infrastructure (road, rail, airport, port) Digital infrastructure	Set clear mandates, funding and guidelines for provision of urban infrastructure and services alongside industrial services, in collaboration with State Urban/ Local Government Department Digital infrastructure is vital for business processes

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

Table 6.2. City typologies: characteristics and available options (cont.)

	City regions	Mid to large cities	Small cities	Transitional areas	Industrial urbanization
Leveraging infrastructure finance	Tax revenue (property, professional, entertainment) and non-tax revenues (user charges, rental income, advertisement fees) Leveraging land assets through FSI premium, Development Charges, Betterment Levies, land lease, etc. Municipal Bonds, Blended Finance, PPPs and private capital, loans from MDBs. CFC Grants (e.g., new cities incubation grant), Centrally Sponsored Schemes (e.g., Smart Cities Mission); SFC, assigned revenue, state grants.	Tax revenue (property, professional, entertainment) and non-tax revenues (user charges, rental income, advertisement fees) FSI premium, Development Charges, lease of land Borrowing from scheduled commercial banks or State intermediaries, Municipal Bonds. CFC Grants, Centrally Sponsored Schemes; SFC, assigned revenue, state grants.	Tax revenue (property, professional, entertainment) and non-tax revenues (user charges, rental income, advertisement fees) CFC Grants, Centrally Sponsored Schemes; SFC, sharing of specific revenues, state grants. Mix of grants and loans from State intermediaries (e.g., TNUDF); grants from MDBs (loan to Central and/or State govt.); CSR Grants.	Property tax and rental income, advertisement fees CFC Grants, Centrally Sponsored Schemes; SFC, sharing of specific revenues, state grants. Grants from State intermediaries or MDBs; CSR Grants.	User charges from services provided to the industries, revenue from development of industrial parks and real estate (residential and commercial) State level grants Foreign investments and private sector financing (e.g., housing).
Access to land	Establish a single land information system, update land cadaster, simplify land conversion PPP Utilizing Development Rights to aggregate built-up spaces Densification through premium FSI Land pooling schemes for the peripheries Negotiated purchase Land lease Eminent domain	Establish a single land information system, update land cadaster, simplify land conversion Fewer options for PPP Densification through premium FSI Land pooling schemes for the peripheries Negotiated purchase Land lease Eminent domain	Update land cadaster, simplify land conversion Negotiated purchase Land lease Eminent domain	Update land cadaster, simplify land conversion Negotiated purchase Land lease Eminent domain	Update land cadaster, simplify land conversion Secure land for industrial park/estate through, i.e., land bank, negotiated purchase, land lease Subsidized serviced land and infrastructure support Eminent domain

VI. ROADMAP FOR IMPLEMENTING THE URBAN GROWTH HUB KNOWLEDGE FRAMEWORK

Table 6.2. City typologies: characteristics and available options (cont.)

	City regions	Mid to large cities	Small cities	Transitional areas	Industrial urbanization
Increase supply of affordable housing	Relax restrictive zoning and development controls Introduce inclusionary zoning, FSI premium and transferable development rights Subsidies and tax incentives through central/state level missions and schemes Leverage underutilized public lands to crowd in private sector affordable housing supply (PPPs) Upgrade informal settlements	Relax restrictive zoning and development controls Introduce inclusionary zoning, FSI premium and transferable development rights Subsidies and tax incentives through central/state level missions and schemes Leverage underutilized public lands to crowd in private sector affordable housing supply (PPPs) Upgrade informal settlements	Focus on integration of planning and infrastructure development Ensure flexible zoning and development controls Introduce inclusionary zoning, FSI premium and transferable development rights Support infrastructure and service connection to cluster of Beneficiary Led Construction schemes Subsidies and tax incentives through central/state level missions and schemes	Focus on basic infrastructure and amenities Subsidies and tax incentives through central/state level missions and schemes	Improve infrastructure (through land pooling, urban upgrading, etc.) around industrial estates to facilitate affordable housing development to attract workers and their families. Incentivize private workers rental housing schemes through PPPs, leveraging public land
Improve connectivity	Presence of a Unified Transport Authority (i.e., CUMTA) Mass Rapid Transit System Multi modal systems and their integration	Bus Rapid Transit System Intermodal integration	Intra and Inter City Buses Shared transport like E-Rickshaws	Shared transport like E-Rickshaws Intercity buses	Provide public transport connecting industrial estate and surrounding urban areas Intercity buses
Invest in urban amenities, livability, and resilience	Develop and implement City Climate Action Plan Develop TOD policy and implementation guidelines Design and develop quality public space to improve walkability, safety and liveability Integrate green infrastructure or nature-based solutions with grey infrastructure Improve access to and quality of education and health services Improve basic infrastructure and services	Develop and implement City Climate Action Plan Develop TOD policy and implementation guidelines Design and develop quality public space to improve walkability, safety and liveability Integrate green infrastructure or nature-based solutions with grey infrastructure Improve access to and quality of education and health services Improve basic infrastructure and services	Design and develop quality public space to improve walkability, safety and liveability Integrate green infrastructure or nature-based solutions with grey infrastructure Improve access to and quality of education and health services Improve basic infrastructure and services	Design and develop quality public space to improve walkability, safety and liveability Integrate green infrastructure or nature-based solutions with grey infrastructure Improve access to and quality of education and health services Improve basic infrastructure and services	Incentivize improving industrial design/ process to ensure worker health and safety, clean air, water and land Improve access to and quality of education and health services Improve basic infrastructure and services

Table 6.2. City typologies: characteristics and available options (cont.)

	City regions	Mid to large cities	Small cities	Transitional areas	Industrial urbanization
Cross-cutting strategy	Promote cluster-based approach to maximize benefits from proximity and collaboration between cities through market linkages, shared infrastructure and knowledge spillovers in a region.				

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ENDNOTES

¹ These categories are aligned with MOHUA's guidelines: [https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I\(2\).pdf](https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I(2).pdf)

² There is much debate about the definition of urban, the extent to which India is urbanizing, and the pace of urbanization. For the purposes of this report, we understand “urban” in terms of how it is defined in the latest available Indian census of 2011, which has three criteria that relate to minimum population, population density, and employment in non-agricultural activities to delineate a settlement as a census town; further, if the settlement has an urban local body administering it, it is considered a statutory town.

³ <https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS?locations=IN>

⁴ Agglomeration economies arise because larger, denser locations make it more likely that workers will get jobs that are a good fit for their skills and needs, that individuals and organizations will exchange ideas and knowledge, and that resources are more efficiently shared—three mechanisms that are more conveniently referred to in the literature as matching, learning, and sharing.

⁵ https://www.indiabudget.gov.in/doc/budget_speech.pdf

⁶ One approach to measuring economic potential can be found in Mark Roberts's “Identifying the Economic Potential of Indian Districts” 2016. An index is constructed comprising five indicators associated with economic potential: market access, transport connectivity, human capital, rate of urbanization, and economic density

⁷ Enablers could comprise a growth coalition of public and private actors, or a major connective infrastructure investment such as an airport or port

⁸ Recognized as urban by the Census of 2011 but under rural administration.

⁹ The number of statutory towns has been increasing based on the urban local body statute accorded by state governments. As per the operational guidelines, 2021 of AMRUT 2.0 (Ministry of Housing and Urban Affairs), the number of statutory towns are 4800.

¹⁰ One of the factors that helps large metros drive growth and innovation is their ability to act as an incubator for many small start-up firms, in addition to their hosting of, e.g., the HQs of larger multinationals. Thus, in addition to the number of firms, the range of firm sizes could also be considered as an indicator of the type of city. How diverse a city's economic base is would also be a good indicator of the type of city it is.

¹¹ These parameters are proposed based on 2011 Census data and the Economic Census 2013-14 and could be further refined based on current data and information. There will be some overlaps in city characteristics across city typologies.

¹² [https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20IIA-IIB\(1\).pdf](https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20IIA-IIB(1).pdf)

¹³ [https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I\(2\).pdf](https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I(2).pdf)

¹⁴ In Delhi for instance, the master plan is revised once in 20 years as per the DDA Act. Thus, land-use changes from industrial to offices/services land use are slow to process. Land prices are high in the core, amalgamation to get larger land tracts is difficult, so new investments choose peripheries.

¹⁵ [Transforming Cities through Transit Oriented Development: the 3V Approach](#). World Bank. 2017.

¹⁶ [https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I\(2\).pdf](https://mohua.gov.in/upload/uploadfiles/files/URDPFI%20Guidelines%20Vol%20I(2).pdf)

¹⁷ There is much debate on the impact of SEZs. See IIHS (2015) for a nuanced discussion

¹⁸ Link to Gujarat Special Investment Region Act. <https://www.gidb.org/acts-policies-sir-act-2009>

¹⁹ The Herfindahl-Hirschman Index (HHI) is used as a measure of concentration, and a lower value indicates a less concentrated (or a more diverse) economy.

²⁰ Odisha rural/urban transition policy: https://urban.odisha.gov.in/node/173346?utm_source

²¹ https://eparlib.nic.in/bitstream/123456789/2963548/1/17_Housing_and_Urban_Affairs_20.pdf

²² World Bank 2023, Technical Assessment, Tamil Nadu Climate Resilient Urban Development Program

²³ <https://www.skillodisha.gov.in/world-skill-centre>

²⁴ Women and Men in the Informal Economy: A Statistical Picture 3rd edition, International Labour Organization (ILO) 2018.

²⁵ About a quarter of all cases decided by the Supreme Court involve land disputes, of which 30 percent concern disputes relating to land acquisition. Source: Wahi, 2019, Centre for Policy Research, Understanding Land Conflicts and Suggestions.

²⁶ These demands translate to about INR 67 trillion, or \$791 billion. Source: Knight Frank, 2024, Affordable Housing in India. <https://www.knightfrank.de/en/research/affordable-housing-in-india-2024-11745.aspx>

²⁷ State governments and urban development authorities should consider developing a land information system that integrates public land records in a single system to deliver a comprehensive data source of public property information.

²⁸ A comparison of FSIs in central business districts of major cities shows that some of the lowest FSIs have been in India—between 1 and 1.5 for Chennai, Delhi, and Mumbai, compared with over 10 in Chicago, Los Angeles, New York, Tokyo, and Singapore (World Bank, 2013). Although there has been relaxation of FSIs in major Indian cities in recent years, analysis of building heights around the world by Barr and Jedwab (2023) indicates that large Indian cities have very few tall buildings. Requirements associated with the extra FSI, such as road width, and challenges amalgamating land into the larger parcels needed to exploit the higher FSI are likely to be key reasons for unrealized FSI.

ENDNOTES

- ²⁹ <https://www.tomtom.com/traffic-index/ranking/>
- ³⁰ Sohaib Athar, Roland White and Harsh Goyal. 2022. "Financing India's urban infrastructure needs: Constraints to commercial financing and prospects for policy action." Washington, D.C.: World Bank.
- ³¹ City Competitiveness Redefined, ARUP
- ³² <https://blogs.worldbank.org/en/sustainablecities/on-urban-heat-india-offers-lessons-to-the-world>
- ³³ Toolkit for Enabling Gender Informed Public Transport and Public Space (World Bank 2023)
- ³⁴ <https://www.wricitiesindia.org/sites/default/files/Delhi-TOD-Policy-Manual.pdf>
- ³⁵ <https://www.maritimegateway.com/chennai-becomes-a-hub-for-warehousing-and-logistics-parks/> <https://www.logisticsinsider.in/chennais-multimodal-logistics-park-construction-kicks-off-in-june-after-years-of-delays/>
- ³⁶ https://mktgdocs.cbre.com/2299/511ce1d0-f1ee-4fa7-9296-769eccc2c527-1059711519/Tamil_Nadu_-_The_Epicentre_of_.pdf
- ³⁷ https://mktgdocs.cbre.com/2299/511ce1d0-f1ee-4fa7-9296-769eccc2c527-1059711519/Tamil_Nadu_-_The_Epicentre_of_.pdf
- ³⁸ <https://www.pmfias.com/chennai-water-crisis/#:~:text=Chennai%2C%20with%20a%20population%20of,a%20persistent%20demand%2Dsupply%20gap.>
- ³⁹ <https://www.pmfias.com/chennai-water-crisis/#:~:text=Chennai%2C%20with%20a%20population%20of,a%20persistent%20demand%2Dsupply%20gap.>
- ⁴⁰ BCP Economic Region can be developed as a cluster of cities
- ⁴¹ Such as sea food processing, garments and textiles, tourism, health and sports
- ⁴² GIS-based Comprehensive Development Plan for Bhubaneswar 2040, p. 88
- ⁴³ <https://tathya.in/world-award-for-jaga-mission/>
- ⁴⁴ <https://indianstates.csis.org/articles/2023-12-11-odisha-approves-the-odisha-rural-urban-transition-policy/odisha-approves-the-odisha-rural-urban-transition-policy/#:~:text=Odisha%20approves%20the%20Odisha%20Rural%2DUrban%20Transition%20Policy%2C%20aimed%20at,growing%20rural%20areas%20adjoining%20cities.>
- ⁴⁵ <https://www.indiaspend.com/how-indore-became-indias-cleanest-city-and-how-others-can-follow/>
- ⁴⁶ <https://www.millenniumpost.in/nation/mps-indore-leading-way-to-green-city-with-innovative-sustainable-solutions-583563>
- ⁴⁷ <https://timesofindia.indiatimes.com/city/bhopal/worlds-largest-600-mw-floating-solar-plant-to-power-omkareshwar/articleshow/112451876.cms>
- ⁴⁸ Indore has to pay an electricity bill of about Rs. 25 crore every month to bring Narmada river water to Indore, 80 kilometres away from Jalud, and for further distribution to every household. The use of electricity generated at the solar power plant at Jalud will reduce IMC's bill by about Rs 5 crore every month <https://www.outlookbusiness.com/planet/sustainability/60-mw-solar-plant-to-power-indores-water-supply-by-year-end>
- ⁴⁹ Over 1,000 start-ups present - <https://timesofindia.indiatimes.com/city/indore/accelerator-prog-tie-ups-with-foreign-varsities-to-boost-startups/articleshow/99627925.cms>
- ⁵⁰ Estimated reduction in cost due to MMLP is presented in Annex 8 (figure A.9)
- ⁵¹ <https://www.hindustantimes.com/india-news/tamil-nadu-government-builds-affordable-housing-for-factory-workers-to-boost-industrial-growth-and-attract-investments-tamilnadu-affordablehousing-industrialgrowth-investment-workers-factories-101682619940182.html>
- ⁵² Cities may also be distinguished on the functions they serve: some may serve as incubators of new products while others focus on the production of mature products (Duranton and Puga, 2001).
- ⁵³ Roberts, Mark. Identifying the Economic Potential of Indian Districts. 2016. World Bank Policy Research Working Paper.
- ⁵⁴ For the often neglected "small towns and census towns", namely, small cities with no specialization, the central government could set up a special program focusing on the top 100 transitional towns for technical and funding support. This option could be further refined to distinguish between transitional towns with greater or less potential for growth. Drawing on the empirical literature on determinants of city growth, this would be those cities with greater market access (i.e., proximity to large cities), greater proximity to an expressway or national highway, a greater share of employment in formal firms, and a greater share of young firms (typically taken to be less than 5 years old).
- ⁵⁵ City GDP estimates are not available, hence proxied by a city's working population.



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